Concurrency and Computation: Practice and Experience

WILEY

REVIEW ARTICLE

Protecting Data in the Cloud: A Systematic Literature Review of Key Management

Shahnawaz Ahmad[1] | Mohd. Nazim[2] | Mohd. Arif[3] | Javed Ahmad[4] | Shabana Mehfuz[5] | Mohd. Aquib Ansari[1]

[1]School of Computer Science Engineering and Technology, Bennett University, Greater Noida, India | [2]Department of CSE, School of Computer Science and IT, Noida Institute of Engineering and Technology, Knowledge Park II, Greater Noida, Uttar Pradesh, India | [3]School of Computer Science and Engineering, Galgotias University, Greater Noida, Uttar Pradesh, India | [4]Department of Computer Science and Engineering, Sharda School of Computing Science and Engineering, Sharda University, Greater Noida, India | [5]Department of Electrical Engineering, Jamia Millia Islamia (A Central University), New Delhi, India

Correspondence: Shahnawaz Ahmad (shahnawaz98976@gmail.com) | Mohd. Aquib Ansari (mansari.aquib@gmail.com)

Received: 24 August 2024 | **Revised:** 10 June 2025 | **Accepted:** 16 July 2025

Keywords: cloud computing | cloud security taxonomy | data security | key management services | privacy-preserving techniques | systematic literature review

ABSTRACT

The rapid adoption of cloud computing (CC) has intensified the demand for robust key management services (KMS) to safeguard data amid escalating privacy and security concerns. Despite KMS being pivotal to securing this increasingly vital technology for modern enterprises, existing literature often lacks a comprehensive, user-centric synthesis of KMS techniques and their practical implications. This systematic literature review (SLR) bridges this gap by analysing 53 significant studies from 2011 to 2024, evaluating KMS in CC with a focus on security, privacy, and operational challenges. Our key contributions include an innovative taxonomy classifying KMS techniques, thoroughly analysing cryptographic methods, and identifying critical research gaps, such as latency reduction and enhanced data privacy. Employing taxonomic classification, meta-analysis, and architectural evaluation, we assessed KMS solutions, including tools like AWS KMS and CloudHSM, revealing substantial progress in KMS architectures and privacy-preserving cryptography. However, persistent challenges, such as latency, scalability, and regulatory compliance, underscore the need for innovative cryptographic and architectural advancements. This survey delivers a comprehensive knowledge base and taxonomy to steer future KMS research in CC, advancing the state of the art by equipping researchers and practitioners with a structured foundation to design more secure, efficient, and compliant cloud systems.

1 | Introduction

Computing evaluates the function $y = f(x)$ based on several inputs, that is, $x1, x2, \ldots xn$, to get the output (y). It is divided into two categories, hard computing and soft computing, depending on the type of input and output. In contrast to soft computing, which deals with ambiguous and inaccurate data, hard computing uses exact inputs and outputs with a systematic approach. Numerous soft computing operations, like fuzzy set theory, rough set theory, etc., have been created explicitly to address imprecision and ambiguity during decision-making. These computing operations can be performed either on the local disk of a computer or on-demand data storage, that is, cloud computing (CC). San Jose introduced CC principles in Search Engine Strategies (SES) in 2006, whereas the National Institute of Standards and Technology (NIST) formally defined the cloud in 2007. Almost every business wants to embrace CC technology, and this is all due to its advantages, including storage, processing, flexibility, and scalability [1]. According to NIST, the classification of CC is shown in Figure 1.

© 2025 John Wiley & Sons Ltd.

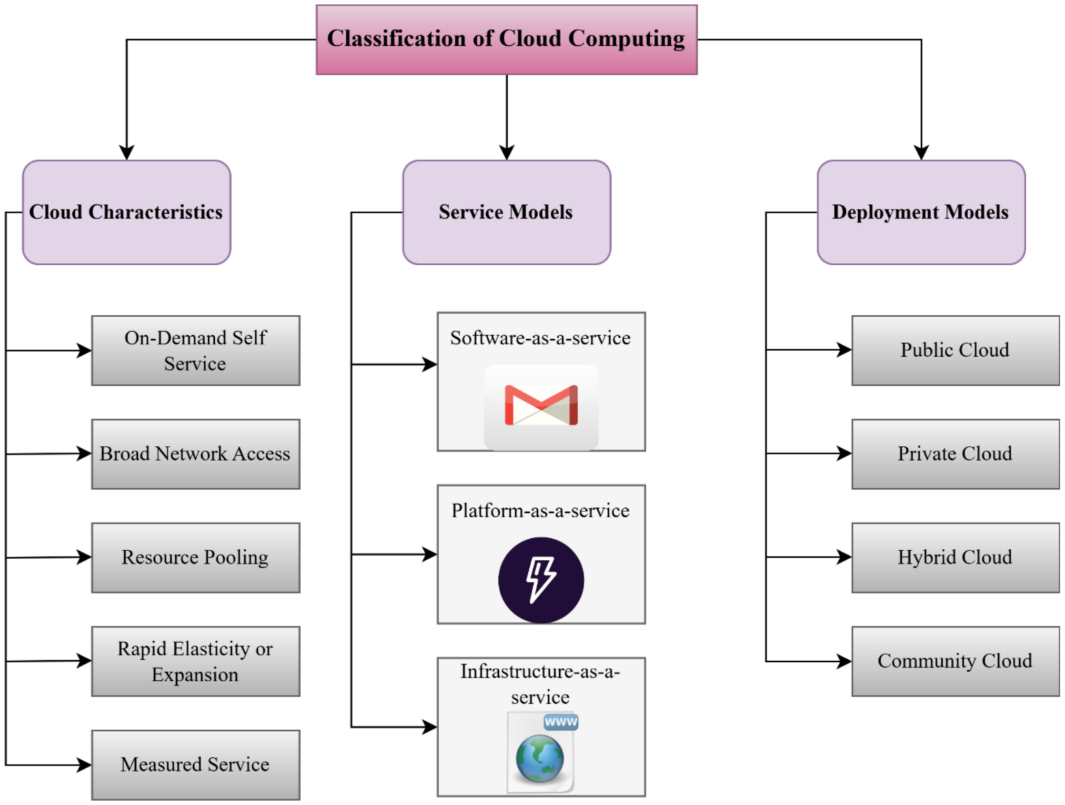


FIGURE 1 | Classification of CC (NIST).

One of the concepts of contemporary computing that is expanding rapidly is CC. Users only pay for the cloud services they utilize; therefore, the initial cost for businesses to adopt CC is relatively low [2]. CC-based technical and service characteristics are much more precisely defined by the processing of heterogeneous data and the provision of computing services (such as storage, databases, networking, software, analytics, computing power, and intelligence) via remote servers located on the internet. Despite the claimed advantages of CC, such as "cheap cost," "high availability," and "flexible application deployment," KMS usage in the real world is still limited [3]. There are various kinds of clouds in literature [4–6]. The three most popular types are public, private, and hybrid cloud deployment models. Cloud deployment models are primarily concerned with the location and management of cloud infrastructure. The concept of the community cloud has not been explored much. The first type is the public cloud, a collection of software (such as an application server or database) that can be used. Infrastructure for one enterprise often makes up a private cloud. The business or a service provider may manage such infrastructure to benefit different consumer groups. The third and last option is a hybrid cloud, which combines private and public CC resources. A community cloud is another option. This cloud shares its computing services and resources with numerous enterprises and can be run by in-house IT staff or outside suppliers [7–9].

This SLR introduces a novel user-centric taxonomy for KMS in CC, synthesizing diverse dimensions such as functionality, cryptographic privacy methods, architectural techniques, and centralized KMS challenges. Unlike prior reviews focusing on technical classifications or specific security aspects, this study bridges enterprise-based outputs, middleware, toolkits, and conceptual frameworks, offering a holistic perspective on KMS evolution and identifying critical gaps in latency reduction and privacy preservation—key concerns for modern cloud adoption.

A few parameters are used to classify them, the primary one being the service levels: Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS), followed by deployment models (public, private, and hybrid clouds), and finally hosting types (external and internal clouds) [10–11]. Infrastructure as a service (IaaS) offers consumers "virtual computers," "storage," "networks," "firewalls," and "load balancers" [12]. With IaaS, users have direct access to the operating system, firewall, virtual machine, and load balancer dashboard, among other components at the bottom of the stack. AWS is one of the largest suppliers of IaaS. The PaaS approach offers its consumers an already-established application platform, saving them time from having to create the infrastructure that supports their applications specifically. One such example of a PaaS provider is Google AppEngine. PaaS solutions frequently feature an API with various capabilities for maintaining and constructing programmable platforms. SaaS, on the contrary, provides ready-to-use internet software solutions. The SaaS software vendor has total control over the application software, which includes "online mail," "project management tools," and "social networking sites." [13]. Researchers have explored CC classification in the cloud environment in greater depth over the past few years, but unfortunately, most of their work was primarily based on academic articles. Many professional articles may welcome this phenomenon because CC is industry-driven. Unlike other works in which the classification of CC is done

passively, this may hamper the ability of the current article to provide a comprehensive picture of the recent developments in this field. Businesses can access enormous amounts of data with little to no upfront expenditure due to the rapidly evolving CC technology [14–16]. Due to security worries, most organizations are still unwilling to migrate their operations to the cloud. Storing data on a cloud server results in the separation of data ownership and management [17]. As a result, the Cloud provider has unrestricted access to the information stored on the cloud server and can access it without the cloud user's permission. Meanwhile, a hacker may steal user data by attacking the cloud server. The cloud platform is also vulnerable to MITM (man-in-the-middle) attacks, data leakage, and other security flaws [18–21]. Information about users is seriously in danger due to the problems mentioned above. The end-user data is at risk of being lost, and problems with data leakage are also conceivable. Data uploaded directly to the cloud can add to concerns like higher bandwidth requirements and substantial latency. These are the most significant roadblocks to the cloud paradigm. Therefore, it is essential to consider it and formulate a suitable response.

Figure 6 displays the volume of consumer fraud and identity theft complaints from 2001 to 2021. In total, 5.74 million complaints of identity theft and fraud were reported to have been filed in 2021. In total, 1.4 million complaints were about identity fraud, 25% were about financial loss, and the rest 2.8 million were related to fraud. It is observed that the volume of complaints increased by 46% from the prior year by 2021. Modern encryption techniques [22–26] have been used to protect user data from overcoming the abovementioned difficulties. Users can use these techniques to encrypt and save their data on cloud servers. In contrast, nowadays, cloud vendors are unable to provide customers with novel methods to safeguard their privacy. Therefore, to address the question, "How can consumers reap the benefits of CC without jeopardizing their data privacy?" Considerable efforts must be made.

Although CC's purported "cheap cost," "high availability," and "flexible application deployment" benefits are lauded, KMS is still not heavily utilized in practice. A variety of types of clouds exist in [. . .]. Most companies are still reluctant to shift their functions to the cloud because of security concerns. Storing information in a cloud server creates a remote data ownership and management separation. Consequently, the Cloud service provider has complete access to all information stored on cloud servers and can independently access the data without the cloud user's consent. In the meantime, a cloud server may be compromised by a hacker who can subsequently steal sensitive information. Active users of the cloud are also exposed to security threats such as MITM, data leakage, and other vulnerabilities. The stated issues are of great concern to the users. Data being exposed in the cloud entails the loss of end-users' valuable sensitive information, and leakage is also probable. Directly uploading data onto the cloud increases existing concerns of additional bandwidth and substantial latency. These barriers are critical to the cloud paradigm. As a result, these need to be addressed calmly, a reliable solution formulated, and investigated thoroughly.

The widespread use of CC for its cost-effective scalability poses a significant challenge in developing effective KMS solutions for data security. User data becomes exposed to security risks, which include unauthorized cloud provider access, data breaches, and man-in-the-middle attacks, because cloud environments have separate data ownership structures from management functions. The existing KMS methods have problems with scalability, latency, and privacy protection, which stop their practical implementation. The SLR research investigates KMS design for CC to combat security hazards through effective performance and privacy protection despite existing solution limitations and outlines future research directions.

The main contribution of the paper is structured as follows:

- Develop a comprehensive, open-access dataset of research on KMS in CC to support academic collaboration.
- Characterize and classify existing KMS solutions using a taxonomy highlighting their similarities and differences.
- Identify and analyse influential research that has shaped KMS in CC.
- Highlight gaps in current KMS research, focusing on latency reduction and data privacy challenges.

This survey paper is structured as follows: In Section 2, we have assessed our findings after conducting a comprehensive literature survey, followed by Section 3, where we give insight into the methodology employed while writing this survey paper. In Section 4, we outlined the studies that have been chosen. Section 5 evaluates the analysis regarding uses, advantages, and gaps identified in various KMS schemes and cryptographic algorithms. In Section 6, we have extensively discussed all our findings and drawn up our concluding remarks. Additionally, we have outlined open research areas for additional study and potential future developments in the CC field in Section 7. Finally, a list of references used for this systematic survey has been provided.

2 | Related Previous Work

Researchers have previously proposed and implemented numerous encryption, decoding, data security, secure transmission, and privacy protection strategies. The vulnerabilities to different types of memory-based attacks, including physical attacks, faulty binding checks, buffer overflows, and others, were discovered by X. Huang and R. Chen [27]. In a cloud platform, KM is typically split into two phases. The first is sharing client keys with cloud providers, which clearly violates security and legal standards. The second approach involves keeping customer keys in a security infrastructure owned by the customer. This is a secure approach, but it reduces performance because there is more delay between customers and service providers. Traditional KM approaches rely on Hardware Security Module (HSM) or Trusted Platform Module (TPM), VMware [28], and Cloud Platforms (AWS and Microsoft Azure) with HSM to make it affordable to every customer while providing reliable security. Still, they are expensive and do not scale well. The security of the key is guaranteed by KMS, such as AWS CloudHSM and Keyless Secure Sockets Layer (SSL). However, an execution debasement is brought on by an interruption between the administrator and the content delivery network (CDN) node. Therefore, looking for a solution that can satisfy scalability and security is necessary.

By identifying the major problems, such as how security can be ensured so that the SSL private key and the content can be stored on unreliable public clouds and how CDN nodes can be enabled to deliver autonomous and efficient data exchange over HTTPS encrypted connections with potential SSL acceleration for enhanced results.

Xiaokang Hu et al. [29] were inspired to develop a novel system for KM called STYX. Furthermore, they examined the difficulty of providing high execution and greater flexibility in SSL/TLS-secured cloud-based CDN systems. Also, data leakage threats like the Heartbleed flaw prevent the adaptable CDN framework from caching the SSL private key on insecure CDN hubs of the cloud platform, further crippling the adaptability and severely degrading performance [30]. Using some of Intel's reliable computing tools, STYX can "transmit the trusted data via untrusted channels and store it on untrusted systems." STYX also enables cloud subscribers to take advantage of CDN nodes. This helps to achieve HTTPS association services with QuickAssist Technology (QAT) acceleration.

A cloud-scale encryption scheme is described by Mathew et al. [31]. It looks at the constraints that influenced the design of AWS KMS and the difficulties that emerge from using a standard mechanism of operation, like AES-GCM (Advanced Encryption Standard Galois Counter Mode), whereas securely supporting enormous amounts of scrambled data that are (at the same time) created and consumed by a tremendous number of clients using different keys.

According to Mathew et al., a modern derived-key mode is designed for the multi-user, multi-key scenario typical at the cloud scale. The need to avoid initialization vector (IV) reuse is activated by being accustomed to common ciphers and modes. As a result, many master key changes are required before approaching initialization vector (IV) collision probability thresholds. Usually, these key changes are too expensive and heavy to deploy in the practical world. As a result, they force the AWS KMS arrangement to employ a cutting-edge style of operation that we refer to as CES-GCM. It is unsustainable for a cloud service provider to incorporate separate customer KM solutions. This paper details the deployment and coordination of the CES framework with 34 AWS services.

To create a Secure Collaborative Key Management system (SCKMS) for mobile cloud data storage that can achieve both data and key security and clarify key escrow and key exposure challenges, Shakkeera et al. [32] proposed a framework. The main goal of the proposed system is to

- Boost information integrity and data security.
- Reduce the computational and storage overhead for encryption and decoding in mobile client devices.
- Minimize how much energy is used by mobile devices.

K. V. Pradeep et al. [33] proposed a novel scheme for transferring a record that will ensure security (CIA) by using an asymmetric key and disseminating it inside the cloud infrastructure with the help of a reliable third party, all while avoiding the introduction of the key and framework. The record, shared by many users and secured from intrusion inside and outside, is vital to the proposed scheme and is secured by CKMS. The information accessed or transferred via any machine present in the cloud system is not secure because it is susceptible to attacks like Identity Access Management (IAM) or capturing an account by internal or external intruders. The security of information inside the cloud environment is remarkably secured using cryptography. Therefore, a standard encryption/decryption component is required, and the key is necessary to ensure the safety of the information stored in the cloud. Each cloud provider has unique security measures in place to protect the key. Even though the service provider always has full access to data and keys, the client cannot fully trust them.

The RSA security level is secure when the N-bit modulus ≥ 2048 bits. However, as a result, the number of erroneous keys also increases. Chandrasegar Thirumalai et al. [34] suggested an Efficient and Non-sharable Public Key Exponent Secure Scheme (ENPKESS) to address this problem. The described methodology is suitable for secure cloud infrastructures on the Internet of Things (IoT). They encrypted ENPKESS keys with Knapsack to attain higher cloud security. Additionally, the authors compared the key generation, encryption, and decryption performance of conventional RSA, ESRKGS, and ENPKESS by varying the N-bit moduli size up to 10 K bits. On comparing the time similarity, RSA and ENPKESS seem to be completely related, but ESR and RSA appear to be only 90% similar. This makes the proposed system extremely resilient to side-channel assault [35].

Secure multiparty computation (MPC) [36–40] allows many servers to collaborate and solve encrypted data issues. As a result, security, including data secrecy, is ensured even when certain servers aggregate their data for vicious purposes. MPC has advanced to the point where it is now viable, at least for a limited range of applications, even though it is still a resource-intensive technique [41].

In many aspects, fully homomorphic encryption (FHE) [42–43] is cloud security's "holy grail" since it allows general computations on encrypted data. FHE enables a cloud subscriber to outsource computing to multiple cloud servers without compromising secrecy, even if all cloud servers are hostile. It is possible to guarantee the accuracy of the outsourced computation when FHE is supplemented with additional methods [44–46]. Additionally, current research [47] investigates outsourcing calculations that require input from several parties. FHE can, therefore, be utilized to safeguard almost every internet computing scenario, including the ones we are looking into. Despite recent developments [48–49], FHE's performance is still insufficient for real-world use. Our methods thus give a more feasible solution to cloud security, at least temporarily, perhaps in combination with MPC during online periods.

Some researchers [50–52] look at how data can be distributed securely among several servers without requiring secret keys to be kept on any clients. These results include secret sharing, error correcting codes, Rabin's information dispersal method [53–54], and other cryptographic techniques to guarantee the stored data's secrecy, integrity, and availability. In addition, forward-security threshold encryption [55] could be used to encrypt files after

shutdown. Therefore, these protocols could be used in our autonomous server concept to protect data during the offline phases.

A method for cooperative KM in CP-ABE is provided in [56]. The current architecture considers distributed generation, private key issuance and storage, and distributed generation. A well-designed and quick attribute revocation is offered to update the keys. A crucial escrow issue and a significant vulnerability are resolved through cooperation. In contrast, it helps in significantly reducing client decryption overhead. The proposed system outperforms the rest of the equivalent CP-ABE strategies in cloud-based outsourced data exchange on mobile devices. Lastly, the security proof of the recommended protocol has been provided.

According to [57], revocation entails denying a user's access. Within the Private Domain, EKMI uses DKPABE (Decentralized Key Policy Attribute-based Encryption) with user revocation and MACPABE (Multi-Authority Cipher-text Policy Attribute-Based Encryption) with attribute revocation [58]. Lazy revocation is used to reduce the processing expenses of the cloud server.

The authors in [59] present a breakthrough encryption method that uses two distant sophisticated non-linear units to generate an encryption key and synchronize via a chaotic driver. The proposed method is sufficiently generic for photonic, optoelectronic, or electronic platforms. Modifying how chaotic signals are generated into the key bit stream is possible. The generated bit series satisfies the National Institute of Standards test suite's criteria for randomization (established by a deterministic procedure).

A LoRaWAN 1.1 root key updating technique was suggested by [60] to evaluate the security and improve LoRaWAN security. The recommended method makes cryptanalysis of security keys in LoRaWAN more challenging. Review and simulation show that the suggested root key update technique uses fewer computer resources than the other key derivation systems, including the one utilized in the LoRaWAN session key update. In addition, the results reveal that the scheme's key has higher unpredictability, which is necessary for a security key.

Hyunsoo Kwon et al. [61] offer a de-duplication approach for practical convergent KM.

The approach has issues with scalability and KM security. A unique safe de-duplication method based on pairing-based cryptography is suggested to overcome these issues because it offers scalable and reliable KM. Moreover, like earlier techniques, it does not require using additional secure channels to transfer essential components and provide secure KM.

The authors in [62] prevent unauthorized parties from accessing the data; emphasis has been placed on the importance of security when transferring data between or inside clouds. Encryption and decryption keys are used to send this type of data. These keys are managed by KLMs (Key Lifecycle Managers), which communicate with one another using a standardized protocol. As a result, cloud platforms must offer a straightforward protocol that enables interaction with third-party key lifecycle managers and allows end users to select a trustworthy KLM.

KM in ransomware is acknowledged and described by Pranshu Bajpai et al. as a prerequisite for identifying flaws that might be used defensively. The maturity of ransomware tested on about 25 samples has been used to illustrate the advancement of KM [63]. Based on research, a ransomware taxonomy has been developed that is comparable to hurricane ratings: Category 5 ransomware is sufficiently cryptographically spreading, but Category 3 ransomware is not [64].

Ahmed Elhadad proposes a system that relies on the re-encryption of DNA proxies. Initial key generation produces three keys: one for the owner, one for the proxy, and one for the user requesting data access. The owner then publishes the encoded material to the cloud using his key. If the user requests access to the data, the proxy will re-encrypt it using a second key issued specifically for the proxy. The user can finally decode data that has been re-encrypted using a third key they generate [65].

A group KM strategy based on PUF's design has been suggested by Hongxiang Gu et al. (physically unclonable function). MIPUF (multistage interconnected PUF) is utilized in an environment with limited energy resources to protect group communications. The MIPUF design can securely distribute, store, and rekey keys efficiently. Furthermore, the design has been demonstrated to be resistant to several attacks. The results reveal that it uses 47.33% less energy globally than the average advanced KMS based on elliptic curve cryptography [66].

Saad Fehis et al. recommend using the Trusted Platform Module (TPM/vTPM) to deliver CKMS as a dependable cloud security service. The existing system leverages TPM's capabilities as a safe and credible option for these kinds of applications. As a result, this example includes TPM's key generation element as a dependable choice for clients concerned about establishing and signing encryption keys using CKMS [67].

Majid R. Alshammari has suggested a practical and safe key distribution system that is feasible, simple to use, and efficient on wireless sensor nodes with limited resources. The work is examined using simulations and hardware implementations. Its effectiveness is then assessed against existing solutions using metrics like "storage cost," "energy usage," "key availability," "MITM attack," "replay attack," and "robustness to node capture attack". The sophisticated encryption standard (AES-128), used to check and encrypt the data being delivered, and the radio wave utilized for communications were both found [68].

Rafael Dowsley and colleagues presented a novel distributed cloud KM method. Data on a public cloud service is encrypted using a different trusted adapter before being stored elsewhere. The encryption key is not permanently stored on the adapter. Instead, parts of the key that are shared among many entities are analyzed and momentarily kept at the adaptor as needed [69].

A strategy for guaranteeing information security and protection for outsourced data kept in cloud storage has been put forth by Vairaprakash Gurusamy et al. It contains KM and a cryptographic tree for protecting outsourced data. The emphasis is on ensuring the security of data that is deployed to the cloud, accessed by numerous authorized parties, and coordinated among them. This

strategy also permits many legitimate parties to outsource their data to the cloud while guaranteeing complete security [70–71].

Farah I. Kandah and colleagues recommended the CSC (Centralized Stateful Connection) method as a useful KM technique in dynamic sensor networks. The suggested solutions provide a good combination of security and efficiency by using public key encryption at the start of a node's existence in a wireless network and transitioning to symmetric encryption for the remainder of the transmission. Sensor networks require a key authentication scheme for key distribution while ensuring that the same key is given to both communicating devices, which are particularly well suited for symmetric cryptography [72].

Adnan Shafiq et al. have offered a technique for absolute out-of-band verification that involves obtaining PIN codes from registered cell phones to complete financial transactions. According to the study, a device that can execute the AES-256 algorithm with user-controlled keys on any PC or laptop can be used to run out-of-band encryption of essential or sensitive data. First, sensitive data is encrypted using the laptop/device, and the encrypted information is sent back to the laptop/computer. Finally, the data is decrypted with the assistance of the same machine. The KMS73 handles the secure distribution of keys.

Li et al. [73] created a new two-factor authentication key contract system in 2017 using biometrics and a password. The protocol uses the user's biometric info and password to enable a safe connection without the necessity of a clever card. This method addressed several security problems that might occur when smart cards are misplaced. Although the technique may withstand insider assaults and the theft of verifiers, we discovered that user anonymity cannot be attained. Additionally, their approach does not include a phase for updating passwords.

Ge et al. [74] have recommended security and exploration techniques for encoded cloud data. The keyword is also automatically updated throughout the sharing phase without the PKG's assistance in their system. They developed the technique using bilinear pairing, symmetrical encoding, and decoding. To ensure that their system was safe against assaults based on keywords and cipher text, they employed a random oracle model.

Irshad et al. [75] advocated UAP for MCE, the verification procedure created for their protocol without the involvement of a third party. Their methodology is computationally intensive due to bilinear pairing processes. "Their protocol doesn't include formal security verification or revocation." Liu et al. created the biometric UAP for MCE [76]. The algorithm was created using ECC and a hash function. Because of this, their procedure does not offer "formal security verification" and does not guarantee the confidentiality of user, server, or log data.

Vivekanandan et al. [77] presented the bilinear pairing-based User Authentication Protocol (UAP) for Mobile Cloud Environments (MCE). They adopted a three-factor verification plan incorporating user individuality, biometrics, and a smart card/mobile device for cloud service admission. A hash purpose, symmetrical encoding, decoding, and bilinear trimming are all components of their method. They used CSPs as part of their protocol to authenticate other CSPs, with the Remote Access Client taking care of MU and CSP registration. The proposed protocol is extremely computationally intensive, and there is no formal demonstration of its security. The system used RAC-based registration and authentication between mobile users and cloud service providers.

Yi et al. [78] suggested a fault-tolerance conference key arrangement based on identity. They drew inspiration from the symmetrical imperfect block design and applied it to the key agreement protocol to simplify communication and computation (SBIBD). To our knowledge, the SBIBD's key agreement protocols are brand new and unique.

Jang et al. [79] calculated a "hybrid security scheme that uses both heterogeneous cryptosystems, such as symmetric and asymmetric (public) key cryptographic techniques." Inappropriately, their arrangement fails to uphold "anonymity property" and does not provide "dynamic controller node addition and medical device addition."

Cryptography-based on public keys, A Mutual Authentication and Key Agreement (MAKA) protocol with proven security and anonymity was proposed by He et al. [80] for multi-server setups. The protocols that have been detailed up to now are password-based. However, such protocols are not safe when subjected to an offline assault that involves guessing the password. An analysis of this protocol has been presented further in this paper in another section titled "Security Comparisons and Cryptanalysis."

Lu et al. [81] presented a dynamic KM approach (LBSs). Different time slots and session keys are used to partition the LBS session into segments. After that, a mechanism for validating vehicular data is provided using probabilistic verification. The method is used to identify nefarious conduct. Additionally, to prevent the calculation interruption, the group signature for the certificate revocation list (CRL) should be checked using the hash message authentication code (HMAC).

Gai et al. [82] discussed the performance and differences of blockchain-based cloud service infrastructure from both a software and hardware standpoint. A detailed analysis of blockchain attacks and existing defenses was provided. An overview of the concept, requirements, structure, and obstacles surrounding the research combining blockchain and edge calculation was done in the paper [83–84]. Very few surveys or taxonomies specifically address blockchain-based trust explanations for cloud computation platforms. As a result, this survey takes a different approach and emphasizes blockchain-based service/resource management in cloud systems.

The mentioned papers cover a variety of topics concerning KMS within CC and technology, which are summarized in Table 1. These include ransomware defense mechanisms, decentralized KMS based on SGX and blockchain, hybrid encryption/multi-cloud task scheduling, performance assessment of cloud KMS providers, privatized access control, and blockchain and lightweight cryptography-based KMS for IoT. It also lists the authors, title, year of publication, area of

TABLE 1 | Summary of related work on key management in cloud and related environments.

References	Year	Focus area	Key contribution	Methodology	Strengths	Limitations	Differentiation from our survey
[85]	2023	Ransomware Key Management	Proposes a framework for ransomware response	Taxonomy-based analysis, empirical study	Detailed ransomware classification	Limited to ransomware, not general KMS	Focused on ransomware only, our survey encompasses broader threat models and privacy frameworks
[86]	2024	Decentralized KMS	Secure data-sharing with decentralized trust	Decentralized architecture, provable security	Enhanced privacy and trust	Scalability is not fully addressed	Specific to decentralized KMS, our survey includes comparative analysis across centralized, decentralized, and hybrid systems
[87]	2023	Decentralized KMS	Intel SGX-based secure KMS	SGX enclaves	Hardware-isolated high security	Hardware-specific dependency	Hardware-centric; our work evaluates broader KMS architecture options, including software and hybrid approaches
[35]	2023	Hybrid Encryption and KMS	Combines AES-OTP, RSA with adaptive KMS	Hybrid cryptography	Strong access control	High computational overhead	Limited focus on hybrid encryption; our survey categorizes multiple cryptographic models under unified taxonomy
[83]	2024	Blockchain-based KMS	Blockchain + digital twin model	Blockchain and digital twin integration	Tamper resistance	Implementation complexity	Focused on one implementation; our survey provides comparative taxonomy of blockchain and other models
[88]	2024	Multi-Cloud Task Scheduling	Uses LLMs and RL for cloud task scheduling	Reinforcement learning	Efficient task scheduling	Not directly KMS-focused	Indirect relation to KMS; our survey maintains exclusive focus on key management and related security
[89]	2023	Cloud KMS Performance	Performance analysis of KMS across providers	Empirical performance testing	Practical cloud insights	Narrow provider scope	Performance-centric; our survey includes architectural and cryptographic perspectives beyond performance
[90]	2025	Privacy-Preserving KMS	Access control with privacy integration	Cryptographic access control	Balance of privacy and efficiency	Use-case specific	Narrow focus; our survey offers a privacy-preserving comparison across multiple KMS models and platforms
[91]	2025	IoT and Blockchain KMS	Group KMS for IoT using blockchain	Multi-layer blockchain	Fault tolerance and scalability	IoT-focused	IoT-centric; our survey covers general cloud KMS, including but not limited to IoT
[92]	2025	Blockchain KMS and Deduplication	Privacy-preserving deduplication via blockchain	Deduplication, blockchain	Data integrity and privacy	High computational load	Data deduplication-specific; our survey includes a broader perspective on data privacy and performance trade-offs
[93]	2025	IoT and WSN KMS	Cost-effective KMS for wireless sensor networks	Lightweight cryptography	Secure and efficient	WSN-oriented, limited to IoT-cloud integration	WSN-oriented and limited to IoT-cloud integration; our survey covers general cloud KMS, including but not limited to IoT, and broader integrations

focus, major contributions, methods used, innovative aspects, shortcomings, and DOI, which show the scope of research in the field of key management.

The following research gaps have been identified based on the literature review:

- Less protection and privacy in outsourced data.
- Data leakage hazards during transmission.
- The effectiveness of privacy-protection methods must be improved.
- During the data transfer, computational and communication expenses must be decreased.
- A variety of user-based security techniques is necessary.

Various SLRs have been presented earlier to secure data transmission and communication in the cloud environment. However, every existing system performs well in different applications and scenarios. Despite this, no SLR mentions higher levels of data security with lower energy consumption when communicating in a cloud environment. Therefore, this systematic survey also includes an efficient data exchange approach with dual security to cover this gap in Section 5.3.

3 | Survey Methodology

This research seeks to conduct a comprehensive and in-depth assessment of the literature to ascertain the state of the art for KMS in the cloud environment. Researchers from various backgrounds have published hundreds of articles on numerous distinct aspects of KMS in CC, such as "edge computing," "fog computing," "distributed computing," "high-performance computing," "grid computing," "optimization theory" and "financial derivatives." As a result, the language used in the literature varies, as seen from past research [94]. Consequently, an SLR is the most exacting technique for finding all relevant work published by scholars from various fields.

3.1 | Research Objectives

The following are the goals of the research given here.

Obj 1. Create a library of work on the overall issue of KMS in CC and make it available as an open dataset to other academicians.

Obj 2. Determine a specific selection of works that have influenced KMS in CC research.

Obj 3. Characterize existing KMS in CC solutions using a taxonomical framework and explain their similarities and differences.

Obj 4. Create a taxonomy of the recent developments by underlining the adopted approaches and methodologies in the field of KMS in CC.

Obj 5. Identify the gaps in the state-of-the-art KMS in CC that can be distinguished based on the methodologies and challenges of the research.

Obj 6. Develop a method for reducing latency while maintaining data privacy during transmission.

3.2 | Research Questions

The SLR has been performed to identify the problems, possible solutions, and applications of KMS in the cloud. We have formulated the following research questions (RQs) to help us accomplish this objective:

- RQ-1: Motivation for designing KMS in CC.
- RQ-2: What types of attacks do KMS approaches mitigate?
- RQ-3: What Privacy-Preserving approaches based on cryptography are available for KMS in a cloud environment?
- RQ-4: What are the KMS architectural techniques?
- RQ-5: Cloud Key Management Services: Problems and possible solutions.

3.3 | Search and Selection Strategy

KMS is a burgeoning industry. As a result, we adopted a survey strategy that required us to gather related work from all authentic sources empirically. So, we have compiled a knowledge base for research from seven important computer science publications and online databases, including: "ACM DL," "IEEE Explore," "ScienceDirect," "SpringerLink," "Wiley Online Library," "PLOS" and "ArXiv." The works from these sources were acquired utilizing a five-step approach outlined in Section 3.3.2. Aggregate indexing services like Scopus and Web of Science were also used, but they were eventually dismissed because they only contained duplicated works extracted from the original publisher databases.

Search string: ((Cloud OR key management OR key management services OR cloud computing) AND (KMS OR key management in cloud OR automation OR limitations OR weaknesses OR strength OR advantages OR disadvantages) AND (review OR systematic review OR literature review OR systematic literature review OR survey OR journey OR literature mapping OR systematic literature mapping OR state-of-the-art)).

3.3.1 | Phase of Search

In the first phase, we use a set of keywords to query the leading publication portals. The returned results and their metadata (title, abstract, and publishing source) are saved in a local knowledge base for later filtering. Figure 2 displays search string combinations.

As necessary identifiers of work in KM in the cloud, the primary keywords (Pk) (KM and key management in the cloud) were picked [95–98]. The essential identifiers for research in the subject of KMS in the cloud were selected as the second-level keywords (Sk) (KMS, Cross-cloud KMS, Federated KM, multicast KM, Hybrid cloud KMS, Multi-cloud KMS, BYOEK, BYOK cloud

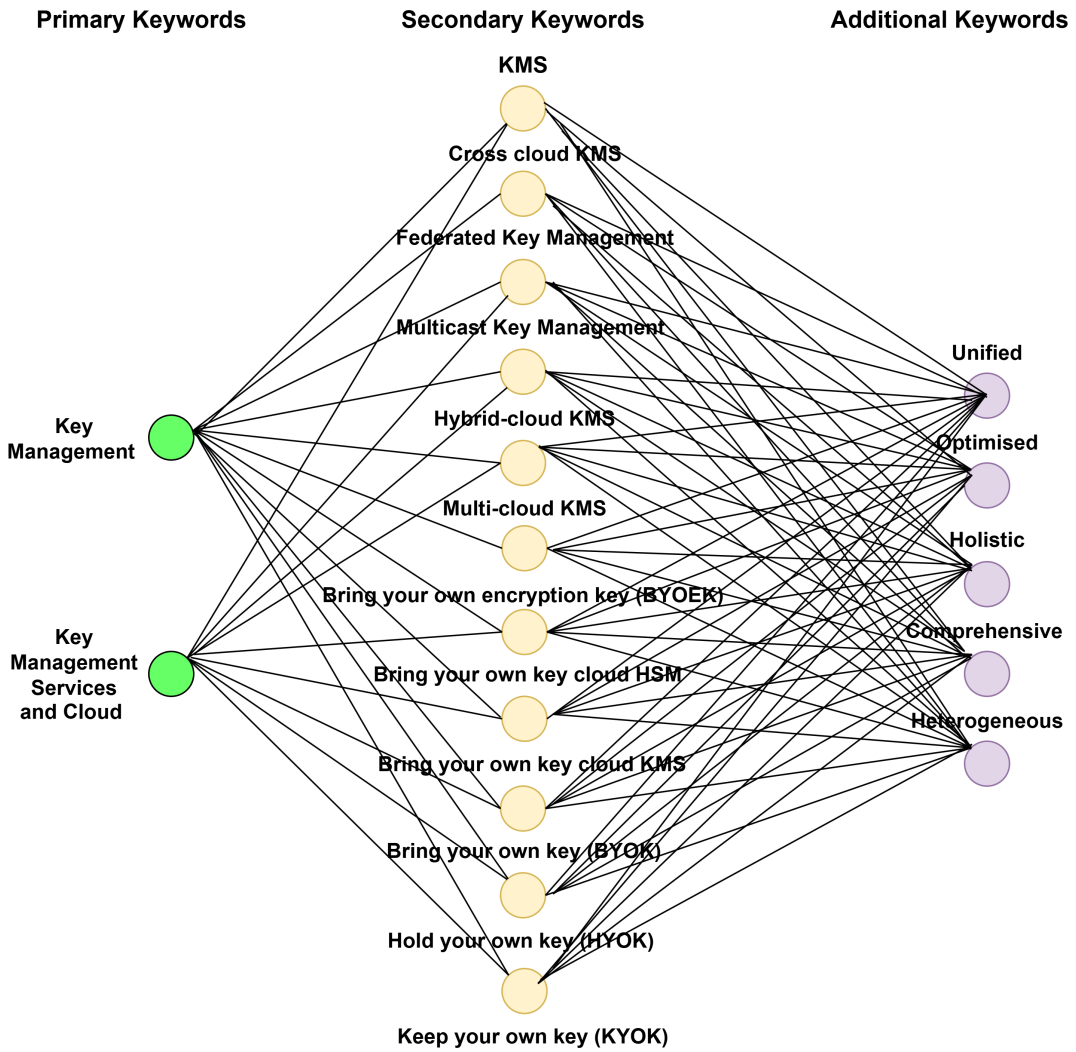


FIGURE 2 | Mapping of primary, secondary, and additional keywords in our knowledge base.

HSM, BYOK cloud KMS, BYOK, HYOK, and KYOK) [99]. The additional keywords (Ak) were identified based on our research (Unified, Optimized, Holistic, Comprehensive, and heterogeneous). Therefore, primary keywords (Pk) were combined with secondary (Sk) or additional keywords (Ak), that is, $\forall Pk \wedge \forall Sk \wedge \forall Ak$.

3.3.2 | Phases of Screening and Criteria for Selection

The knowledge base is filtered using three processes derived from the publication's abstract, introduction, and conclusion sections. We intentionally excluded information about the authors and publishing venues to ensure an unbiased evaluation of articles and avoid favoring publications based on their origin. This allowed us to assess relevance solely based on the article's metadata and key content. We also observed that all titles and abstracts accurately reflect the article's content. However, we concluded that our method was required to eliminate assumptions based on provenance. Figure 3 summarizes the IC (IC_1, IC_2, IC_3) and EC (EC_1, EC_2, EC_3, EC_4) used in the selection process.

Table 2 and Figure 4 provide a brief quantitative summary of our findings from the search, screening, and inspection processes conducted across the seven published libraries. A vast number of papers were included in the survey. However, this number was reduced due to the screening steps, albeit at a far slower rate than anticipated. This is because many publications misrepresent themselves by claiming to offer a comprehensive cloud management solution when they merely provide a supporting technology or model. (Figure 5a–e) shows the phases of an SLR based on different databases.

3.3.3 | Inspection Phase

The accepted articles were carefully examined to determine their significance and contribution to the field of KMS. The primary objective of the research is to build a systematic knowledge base. Hence, the publications were grouped depending on their contributions.

4 | The Review: A Reporting

A detailed summary of the selected studies has been provided in this section.

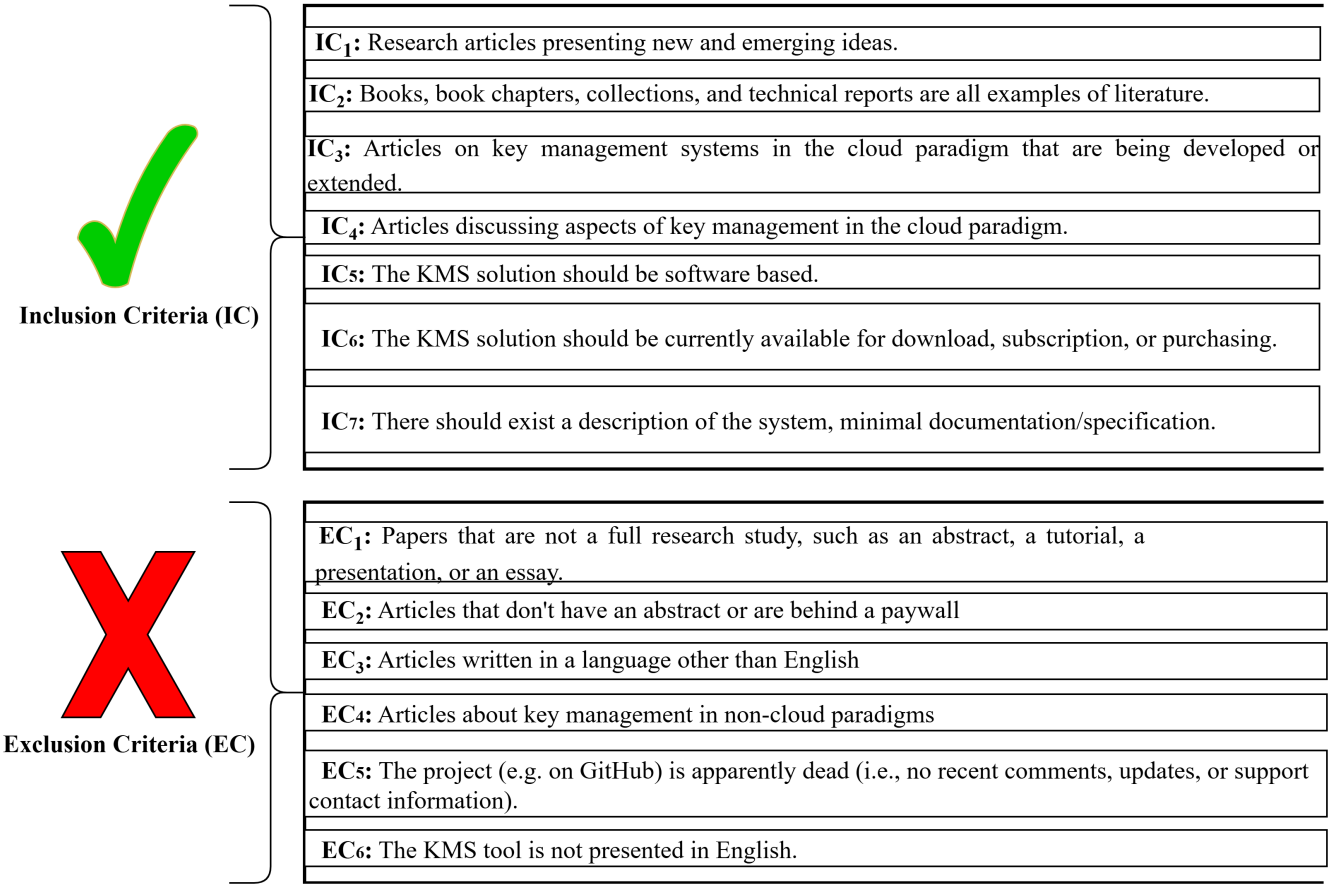


FIGURE 3 | Criteria for study selection based on inclusion criteria (IC_1, IC_2, IC_3) and exclusion criteria (EC_1, EC_2, EC_3, EC_4).

TABLE 2 | Phases of selection and outcomes.

Selection criteria (phases)	ACM DL	arXiv.org	IEEE Explore	PLOS	Science Direct	Springer Link	Wiley Online Library	Total
Search-based on Keywords	4326	87	402	11	3525	1314	195	9860
Screening based on Title	111	21	133	5	167	29	25	491
Screening based on the Abstract	14	8	69	0	72	12	10	185
Screening based on the Introduction and Conclusion	9	6	40	0	35	9	9	108
Inspection based on the Full article	6	1	32	0	10	2	2	53

4.1 | Selection Process Outcome: An Overview

In this section, the SLR's top-level findings have been presented. The outcome of the search, filtering, and inspection procedures across the seven published libraries is succinctly presented in a quantitative format in Figure 5a–e (i.e., search-based on keywords, screening based on title, screening based on abstract, screening based on introduction and conclusion, and inspection based on the full article).

4.2 | Selected Studies

The survey shortlisted 53 noteworthy articles on KM at the cloud scale published between 2011 and 2024. Figure 6 demonstrates that a high proportion of these articles (16) were published in journals, whereas just 32 were published at conferences, and five were published in other publication types.

Elsevier and Springer are the leading distributors of scholarly articles, with platforms like ACM DL, IEEE Explore,

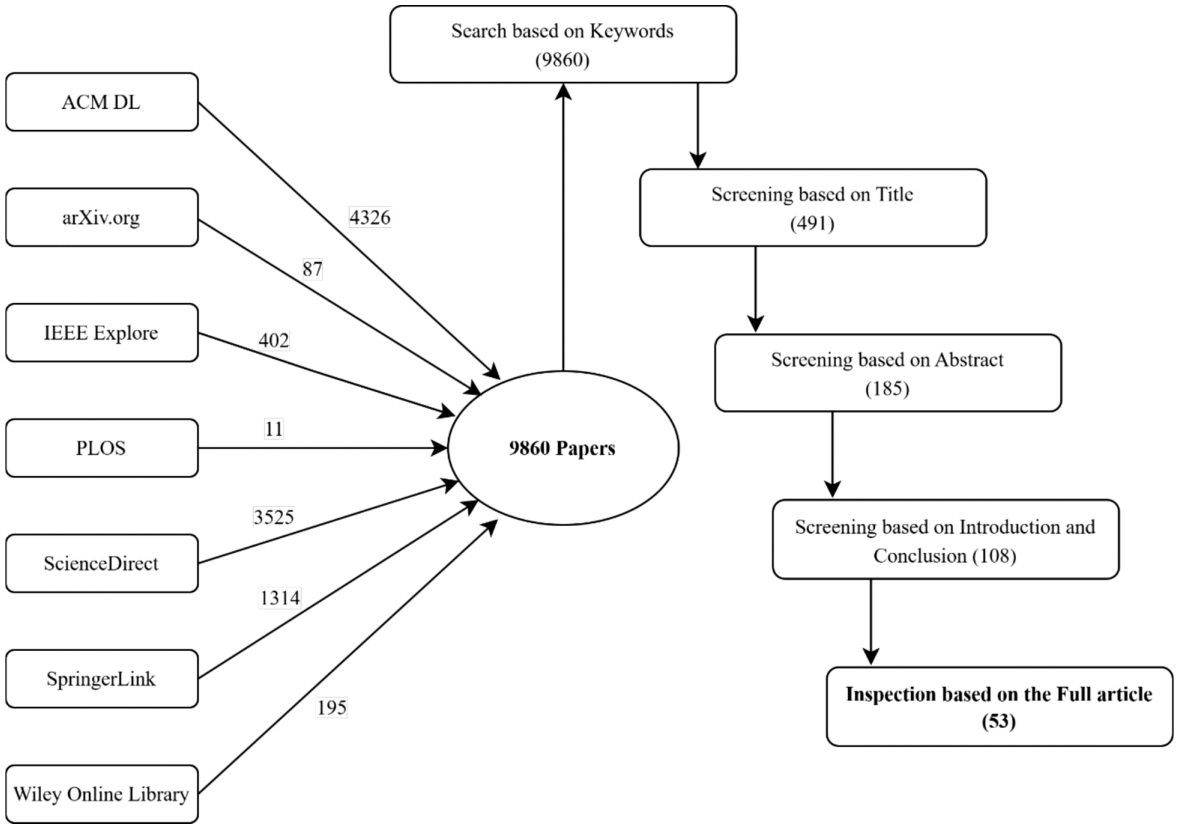


FIGURE 4 | Studies selection process.

ScienceDirect, SpringerLink, Wiley Online Library, PLOS, and ArXiv. Elsevier's prominence is partly attributed to its FGCS journal, which was among the first to solicit research on knowledge management in cloud environments. On the other hand, Springer is known for publishing specialized conference proceedings.

5 | Results of Our Review

The responses to the research questions have been presented in this section.

5.1 | Motives for KMS Design in Cloud Computing (RQ1)

This section discusses the challenges that led to the development of KMS in the cloud and how various publications recognized these issues. Most articles are transparent and straightforward about their motivations. An appropriate KM method is used to achieve secure and trusted communication among cloud-connected devices. As a result, only authorized and trustworthy users in the cloud environment can access authenticated data [100]. This further leads us to the fact that the KMS is critical in ensuring secure and genuine communication between devices.

On the other hand, the KM approaches encountered several obstacles, including increased computation costs, network complexity, longer calculation times, and a variety of other variables. As a result, a strong authentication strategy is required. Hence, this research article offers a novel, safe authentication strategy based on KM for secure communication among cloud devices. The necessity to overcome one or more of 18 fundamental security issues [101–102] has been identified as a motivator for KMS in CC:

- Provides mutual authentication.
- Provides multi-level authentication.
- Requires identity-verification table.
- Server spoofing attack resistance.
- Stolen verifier attack resistance.
- Privileged insider attack resistance.
- Password guessing attack resistance.
- Provides strong user anonymity.
- Known session-specific temporary information attack resistance.
- Impersonation attack resistance.
- Reply attack resistance.
- Man-in-the-middle attack resistance.
- Provision for revocation and re-registration.
- Free from denial-of-service attack.
- Profile table-stolen resistance.

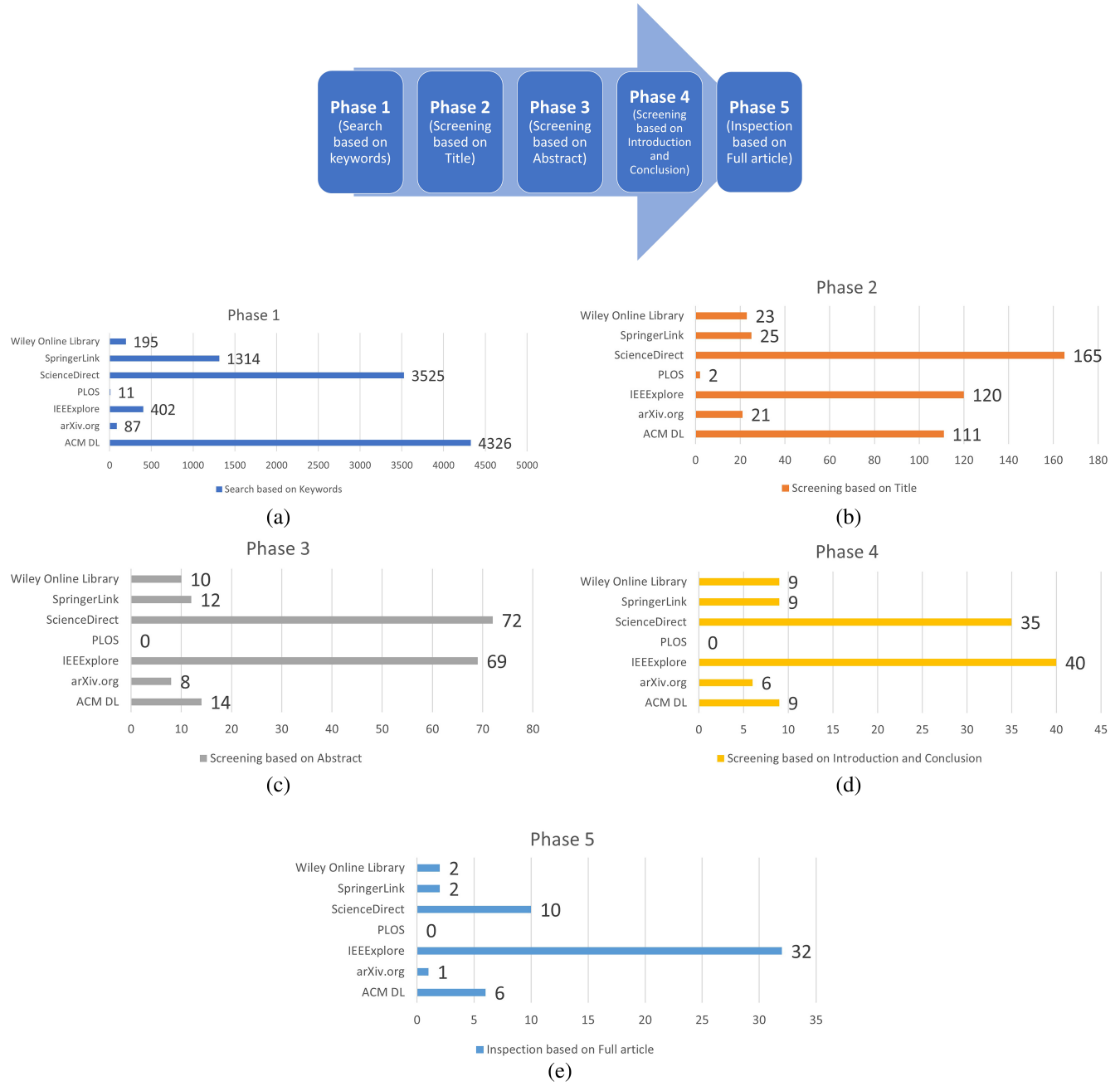


FIGURE 5 | Stages of the systematic literature review. (a) Keyword-based search across multiple knowledge bases. (b) Title-based search across multiple knowledge bases. (c) Abstract-based search across multiple knowledge bases. (d) Search analysis using introduction and conclusion across different knowledge bases. (e) Full-text search across different knowledge bases.

- Key resilience.
- Reconnaissance attack resistance.
- Free from theft attack.

Modern KMS approaches [3, 103–106] were used to protect user data and address the abovementioned difficulties. Utilizing these KMS approaches, users can now encrypt their data and store it on cloud servers. Cloud servers can use it to do calculations after that. On the other hand, cloud servers cannot offer customers fresh perspectives while protecting their privacy. In addition, efforts must be made to address how CC can serve clients without compromising their privacy.

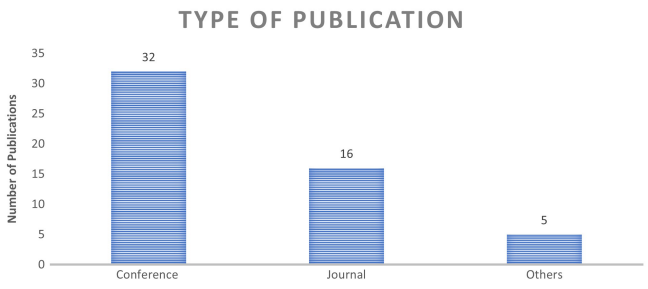


FIGURE 6 | Overall breakdown of selected research publication types.

5.2 | What Types of Attacks Do KMS Approaches Mitigate? (RQ2)

The CC environment is vulnerable to attacks from malicious software, spoofing, viruses, denial-of-service attacks, and other sorts of attacks. Addressing this question will help us comprehend the typical attacks that could harm the cloud platform and categorize KMS solutions depending on the assaults they are intended to mitigate. Cyberattacks can be categorized based on the specific types of cybercrimes they commit. These categories include [107]: Cybertrespass: Activities like traffic analysis, eavesdropping, snooping, password attacks, SQL injection, salami attacks, and data manipulation. Cyberviolence: Attacks that disrupt or deny services, such as denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks. Cyberpeddling: Malicious activities like keylogging, social engineering, and session hijacking. A Cybercrime Matrix categorizes cybercrimes based on the technology employed and the modus operandi. This framework divides cybercrimes into four primary categories [108]: Technology-Driven Crimes: These involve direct attacks on machines, utilizing machines for criminal activities, or exploiting vulnerabilities within machines. Cyber-Assisted Crimes: These crimes leverage technology to facilitate traditional criminal activities, such as social engineering, peer-to-peer fraud, and disseminating harmful information. Cyber-Enabled Crimes: These large-scale frauds are made possible using technology and the internet. Cyber-Dependent Crimes: These crimes are entirely reliant on technology and the digital landscape, including distributed denial-of-service attacks, mass hacking, phishing, ransomware, spam, and hate speech. Alshabibi [109] observes that cloud technologies have fundamentally altered online crime by enhancing computational power, storage capacity, and affordability. This transformation empowers cybercriminals to execute more sophisticated attacks at a lower cost. Wall says in his conclusion that even though cloud technologies are used to help with these purchases, people will always use the internet to source tangible goods. On the other hand, cloud-based cybercrime can include data theft, such as the theft or manipulation of an entire cloud. When the fog is removed, the crime disappears. Cloud-based criminal activities like mass spam would significantly diminish without cloud technologies. For example, the volume of spam could drop from 10 billion emails every 10 s to 10 million every 10 min.

There is no universal solution to cloud-related crimes as new threats and vulnerabilities constantly emerge. However, existing research provides valuable insights into countermeasures and solutions. Studies by Mithunzi et al. [110] and Kumar and Goyal [111], among others, offer a comprehensive analysis of cyber threats and their mitigation strategies. Several studies by researchers, including Asvija et al. [112], Litchfield and Shahzad [113], Senyo et al. [114], De Donno et al. [115], Singh et al. [116], Fernandes et al. [117], Modi et al. [118], Coppolino et al. [119], Khalil et al. [120], Latif et al. [121], Zissis, and Mithunzi have proposed comprehensive approaches to address cloud security challenges. These approaches focus on conducting thorough security analyses and implementing effective countermeasures to safeguard cloud environments.

Table 3 outlines the cloud KMS threats the Cloud Security Alliance (CSA) identified. The CSA regularly publishes updates on emerging risks and changes in threat rankings within the cloud environment [111, 122]. The risks were prioritized in the CSA's 2019 Final Report based on industry expert surveys. These rankings reflect the most significant threats and vulnerabilities associated with CC. The CSA noted that the "new, highly scored items" in the survey indicate a growing consumer sophistication and understanding of cloud risks. These issues are primarily cloud-specific, reflecting a technological landscape where consumers are actively considering cloud adoption. Examples include potential control plane vulnerabilities, meta- and application-structure issues, and limited cloud visibility.

Figure 7 illustrates consumer fraud and identity theft complaints from 2001 to 202 [122]. According to reports, 5.7 million complaints against fraud and identity theft were filed in 2021. Identity theft accounted for 1.4 million complaints, money loss accounted for 25% of instances, and fraud accounted for the remaining 2.8 million. The number of complaints surged by 46% in 2021 compared to the previous year. Identity theft complaints accounted for 25% of all complaints in 2021, surpassing other categories. Imposter scams, a type of fraud, received 984,756 reports (17.2% of all reports). Credit bureaus, information furnishers, and report users were the top three sources of reports to Sentinel, representing 10.3% of all reports. The report categories include: Identity Theft (1,434,676, i.e., 25.01%), Imposter Scams (984,756, i.e., 17.16%), Credit Bureaus, Information Furnishers and Report Users (59,292, i.e., 10.33%), Online Shopping and Negative Review (398,283, i.e., 6.94%), Internet Service (121,445, i.e., 2.12%), Telephone and Mobile Service (92,802 i.e., 1.62%), Privacy, Data Security, and Cyber Threats (70,177 i.e., 1.22%), Credit Card (65,173 i.e., 1.14%), Computer Equipment and Software (15,701 i.e., 0.27%) [122]. Several reports registered by this type of complaint have been depicted in Figure 8.

5.3 | What Privacy-Preserving Approaches Based on Cryptography Are Available for KMS in a Cloud Environment? (RQ3)

Yuan and Yu proposed a multiparty neural network approach for collaborative learning using encrypted data. Based on back-propagation neural networks, their method demonstrated accuracy, efficiency, and security. They employed doubly homomorphic encryption to process encrypted data [123]. They concentrated less on increasing algorithm performance and more on enhancing data processing. A deep computation paradigm for privacy-preserving homomorphic encryption was created by Zhang et al. [124]. They made better use of the cloud's learning capabilities by offloading the expensive procedures. The Taylor theorem was used to implement the exponential process needed in the Sigmoid function. A privacy-preserving method based on the DPHE scheme was presented by Yonetani et al. [125], which successfully trains visual classifiers across distributed private data. The proposed approach gave multiparty protected scalar products for high-dimensional classifiers while minimizing the computational cost. Regardless, one operation (addition or multiplication) can be supported simultaneously. A straightforward approach based on the multi-key

TABLE 3 | Major KMS threats and challenges to CC, according to the CSA.

Threat category	Egregious Eleven (2019)	The Treacherous (2016)	CSA (2010)	CSA (2013)	Security responsibility	Cloud service model
Data Breaches	Breach of data	Data breaches	Abuse and nefarious use of cloud services	Data breaches	Client and CSP	SaaS, PaaS, IaaS
Misconfiguration and Inadequate Change Control	Misconfiguration and Inadequate Change Control	Poor identity, credential, and Access management	Unsafe application programming interfaces	Data loss	Client	SaaS, PaaS, IaaS
Lack of Cloud Security Architecture and Strategy	Lack of Cloud Security Architecture and Strategy	Unsafe APIs	Malicious Insider	Account hijacking	Client	PaaS, IaaS
Insufficient Identity, Credentials, Access, and Key Management	Insufficient Identity, Credentials, Access, and Key Management	System and application vulnerabilities	Shared technology vulnerabilities	Insecure APIs	Client	SaaS, PaaS, IaaS
Account Hijacking	Account Hijacking	Account hijacking	Loss/Leakage of data	Denial of service	Client and CSP	SaaS, PaaS, IaaS
Insider Threat	Insider Threat	Malicious Insider	Hijacking of accounts, services, and traffic	Malicious Insider	Client	SaaS, PaaS, IaaS
Insecure Interfaces and APIs	Insecure Interfaces and APIs	Advanced persistent threats	Unknown risk profile	Abuse of Cloud Services	Client and CSP	SaaS, PaaS, IaaS
Weak Control Plane	Weak Control Plane	Loss of data	—	Insufficient Due Diligence	Client	SaaS, PaaS, IaaS
Meta-structure and Applistructure Failures	Meta-structure and Applistructure Failures	Insufficient due diligence	—	Shared technology issues	Client and CSP	SaaS, PaaS, IaaS
Limited Cloud Usage Visibility	Limited Cloud Usage Visibility	Abusive and dishonest use of cloud services	—	—	Client and CSP	SaaS, PaaS, IaaS
Abuse and Nefarious Use of Cloud Services	Abuse and Nefarious Use of Cloud Services	—	—	—	Client and CSP	SaaS, PaaS, IaaS

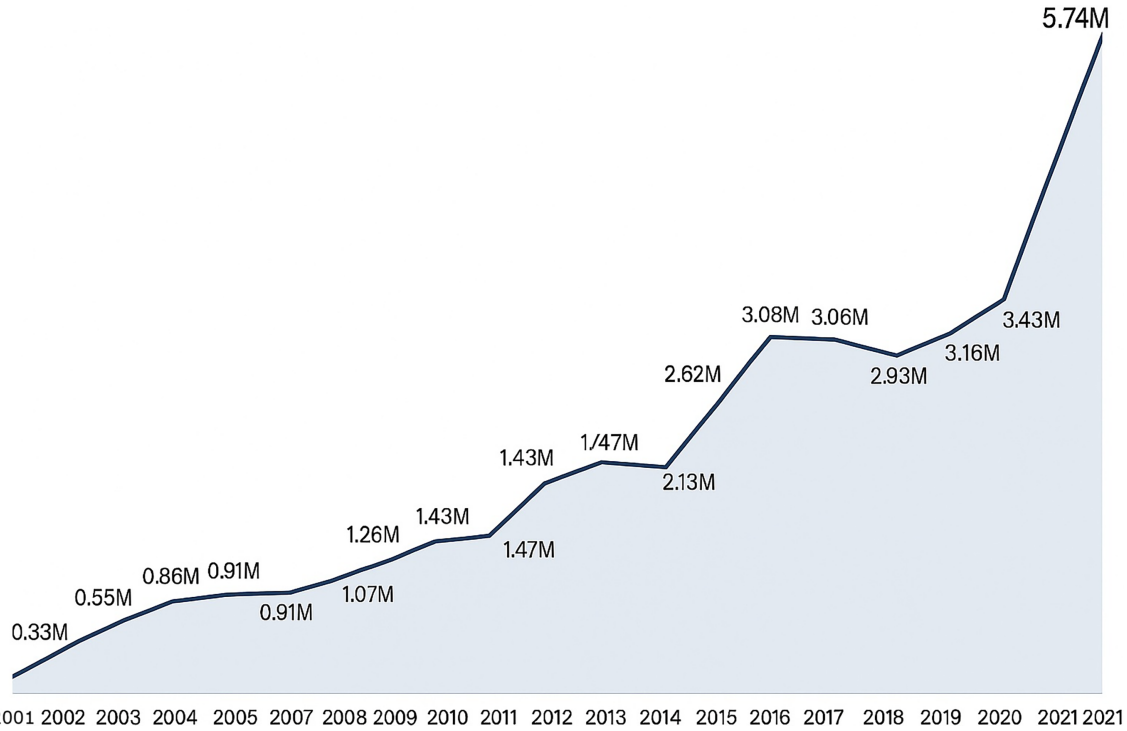


FIGURE 7 | Trends in fraud, identity theft, and other reported crimes [122].

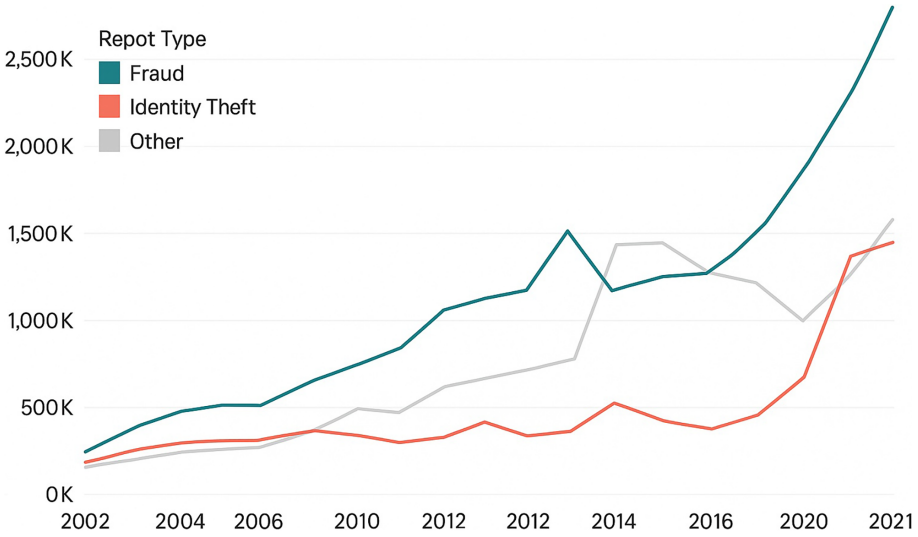


FIGURE 8 | Number of reports by type.

fully homomorphic encryption (MK-FHE) method was introduced in [126]. The researchers created a complex model that uses complete homomorphic encryption and double decryption to learn encrypted content in the cloud (FHE). Unfortunately, this method requires hefty computational and communication costs. A framework for Privacy-preserving Outsourced Classification in CC (POCC) that protects privacy and successfully permits infinite ciphertext multiplication and addition operations was created by [127]. A proxy completely homomorphic encryption approach based on Gentry's method was used to safeguard the data and queries. The proposed architecture did, however, increase the cost of computation and transmission. Ma et al. recommended using a privacy-preserving deep learning model (PDLM) to train the model on data that had been encrypted using the owners' keys [128]. To complete the training confidentially, a suite of computer methods based on stochastic gradient descent (SGD) was used. Despite reducing storage overhead, the model incurs a high computing expenditure and decreased classification accuracy. The authors in [129] outline an outsourced classification method that protects user privacy and offers classification services through encrypted data. Additionally, they developed two practical secure classification protocols: one for the hyperplane decision-based classification and the other for the Naive Bayes classifier. However, the beginning of a categorization inquiry is frequently

TABLE 4 | Summary of the cryptographic and perturbation-based data privacy-preserving technologies.

Sources	Approaches used	Advantages	Disadvantages
[131]	Hashing with a Geographical Constraint	Take care of the service suggestion	Low efficiency
[132]	Algorithm of the Hash-Solomon code	Privacy is well-protected	High cost of computation
[133]	Encryption using public key authentication	Inside the keyword, handle to resist	Single keywords
[123]	BGV encryption is fully and doubly homomorphic.	A secure scalar product that efficiently encrypts data	Low efficiency
[134]	Scheme for Searching by Keywords	For storage and computation, use two cloud servers.	High complexity
[135]	Doubly homomorphic encryption	Multi-party secure scalar product was supported	Multiplication and addition are also possible.
[136]	The accuracy of the stemming algorithm is being improved	At a lower cost, data is outsourced	The keyword weight is not considered
[126]	Encryption with multi-keys that are entirely homomorphic	Protect the privacy of important information.	Low efficiency
[137]	Model TF-IDF	Update the data that has been outsourced at a lower cost	No effect on computing efficiency
[128]	Public-Key Cryptosystem with Two Trapdoors Distributed	More efficiency	Less efficiency
[138]	Dynamic Data Operation, KNN, TF-IDF	Solve the multi-keyword ranked search problem	Doesn't investigate verifying for integrity
[139]	Algorithms for optimization	Data on training should be kept safe	High level of complexity
[140]	The BPNN and Extreme Learning Machine schemes	Partitioned Data can be shared horizontally and vertically with several parties	Communication costs are high
[141]	Encryption i.e., homomorphic	Numbers are randomly split to encrypt data.	Either multiplication or addition
[142]	ElGamal and homomorphic schemes	Data can be partitioned in whatever way you like	Two party
[143]	ElGamal and homomorphic schemes	Data can be partitioned in whatever way you like	Two party
[144]	€-Differential Privacy	Reduce the time it takes for a query to be answered	Less accuracy
[145]	Mechanism of Laplace	Data security	Overhead protection for computation
[146]	Encryption that is homomorphic Privacy Differential	More efficiency	The cost of computation is high
[147]	Gaussian mechanism	High accuracy	There is no protection from classifiers
[148]	€-Differential Privacy	Data privacy must be protected	Forge information
[149]	Laplace mechanism	Achieve a balance between precision and noise	Individual privacy is respected
[150]	€-Differential Privacy w event	Data security	A complicated system
[151]	€-Differential Privacy €-Homomorphic encryption	Cost is low	Data exchange is restricted

influenced by user actions. A privacy-preserving Naive Bayes classifier algorithm was created by Gao et al. [130] to stop information from leaking. The Naive Bayes' privacy was protected via a double-blinding approach. Privacy-preserving data techniques based on perturbation and cryptography have been compiled in Table 4.

TABLE 5 | Key management service (KMS) approaches comparison.

Feature/aspect	AWS KMS	STYX	SCKMS	Google Cloud KMS	Azure Key Vault
Key Management Model	Centralized (AWS-managed)	Decentralized (Distributed Trust)	Scalable, Multi-tenant	Centralized (Google-managed)	Centralized (Azure-managed)
Integration	Seamless with AWS services	Multi-cloud capable	General, designed for scale	Strong with Google Cloud	Strong with Azure services
Security Focus	High, robust multi-key	Enhanced against insider threats	Scalable security	Fine-grained access controls	High compliance, HSM support
Scalability	High, within AWS ecosystem	Depends on implementation	High, designed for multi-tenancy	High, within Google ecosystem	High, within Azure ecosystem
Performance	Potential degradation during key rotation	Higher complexity, potential overhead	Latency in high-frequency transactions	Cost considerations for high usage	Potential latency in large-scale deployments
Ecosystem Lock-in	Strong AWS lock-in	Minimal	Minimal	Strong Google lock-in	Strong Azure lock-in
Use Case Suitability	Enterprise AWS, financial apps	Multi-cloud, critical infrastructure	Mobile cloud, IoT networks	AI/ML cloud, SaaS on GCP	Enterprise SaaS, regulated industries
Key Distribution	AWS-managed	Decentralized	Scalable, multi-tenant	Google-managed	Azure-managed
Hardware Security	AWS HSM options	Software-based	Software-based	Google HSM options	HSM support (optional)
Compliance	AWS compliance standards	Depends on implementation	Depends on implementation	Google compliance standards	High compliance standards

In Table 5, we have compared the following KMS approaches: AWS KMS, STYX, SCKMS, Google Cloud KMS, and Azure Key Vault. The comparison includes the following features: model of key management, integration, security, scalability, performance, ecosystem lock-in, primary usage, key distribution, hardware security, and regulatory compliance. AWS, Google, and Azure provide centralized proprietary, cloud-based services with rich functionality but ecosystem lock-in. In contrast, STYX offers a decentralized multi-cloud KMS that is more protective against insider threats but with greater complexity. The SCKMS system provides multi-tenant key distribution services with scalable capabilities that optimize performance in mobile and IoT networks, as noted in 177. Organizations can use the provided table to make informed KMS decisions by understanding how security measures relate to scalability performance, as well as security concerns involving vendor lock-in.

Table 6 presents a comparative analysis of quantum-resistant cryptographic approaches for integration into Key Management Systems (KMS). Lattice-based cryptography, like Kyber, offers high integration potential with strong security, though it carries moderate to high computational overhead. Hash-based signatures, such as XMSS, are suitable for authentication with robust security and lower computational overhead. Code-based encryption, exemplified by McEliece, provides strong security but suffers from high computational overhead due to large key sizes. Multivariate polynomial schemes like Rainbow have limited KMS applicability and face security vulnerabilities. Despite its lightweight keys, supersingular isogeny-based cryptography (SIKE) has been compromised and removed from NIST's standardization process. The table highlights the trade-offs between security, performance, and integration potential for each approach in the context of future-proofing KMS against quantum computing threats.

5.4 | What Are the KMS Architectural Techniques? (RQ4)

These solutions protect users' cryptographic keys from unauthorized access and misuse by implementing tamper-resistant hardware and software components in one or more layers of the cloud infrastructure. Many of these solutions assume that public cloud providers like Amazon and Microsoft prioritize their reputations and can be trusted, but VM domain managers may pose a threat. This presumption needs to be viewed with a grain of salt because domain administrator positions are regularly found within the organizational structure of cloud providers. Figure 9 illustrates the methods for protecting KMS in a CC environment using cryptographic keys. The architecture also includes a Cloud System and an On-Premises System. Figure 9 on the following page depicts the KMS architecture.

The user inheritance in this design can be considered as two sorts of users (user1, user2, ..., usern): users who generate content and users who receive push notifications of new content.

TABLE 6 | Quantum-resistant cryptographic approaches for KMS integration.

Approach	KMS integration potential	Security features	Computational overhead	Notable references/ standards
Lattice-Based Cryptography (NTRU, Kyber)	High (key encapsulation, encryption)	Strong resistance to quantum attacks, provable security	Moderate to High (large keys, intensive computations)	NIST PQC finalists (Kyber — CRYSTALS) (2022–2024)
Hash-Based Signatures (XMSS, SPHINCS+)	Medium (authentication, digital signing)	Stateless/Stateful variants, strong resistance to classical and quantum attacks	Low to Moderate (slower signing/verification in stateless)	NIST-approved standard (XMSS), SPHINCS+ finalist
Code-Based Encryption (McEliece)	Moderate (secure key exchange)	Longstanding resistance to quantum attacks, high confidence	High (very large public key sizes)	NIST Round 3 finalist — Classic McEliece
Multivariate Polynomial Schemes (Rainbow, GeMSS)	Low to Moderate (limited applicability)	Efficient in signing, smaller key sizes than McEliece	Moderate (Rainbow deprecated due to structural attacks)	Initially NIST PQC Round 3, Rainbow dropped in 2022
Supersingular Isogeny-Based Cryptography (SIKE)	Low (experimental, due to vulnerabilities)	Lightweight keys and encryption (once promising)	Very High (slow performance + SIKE breaks)	SIKE broken (2022), removed from NIST PQC process

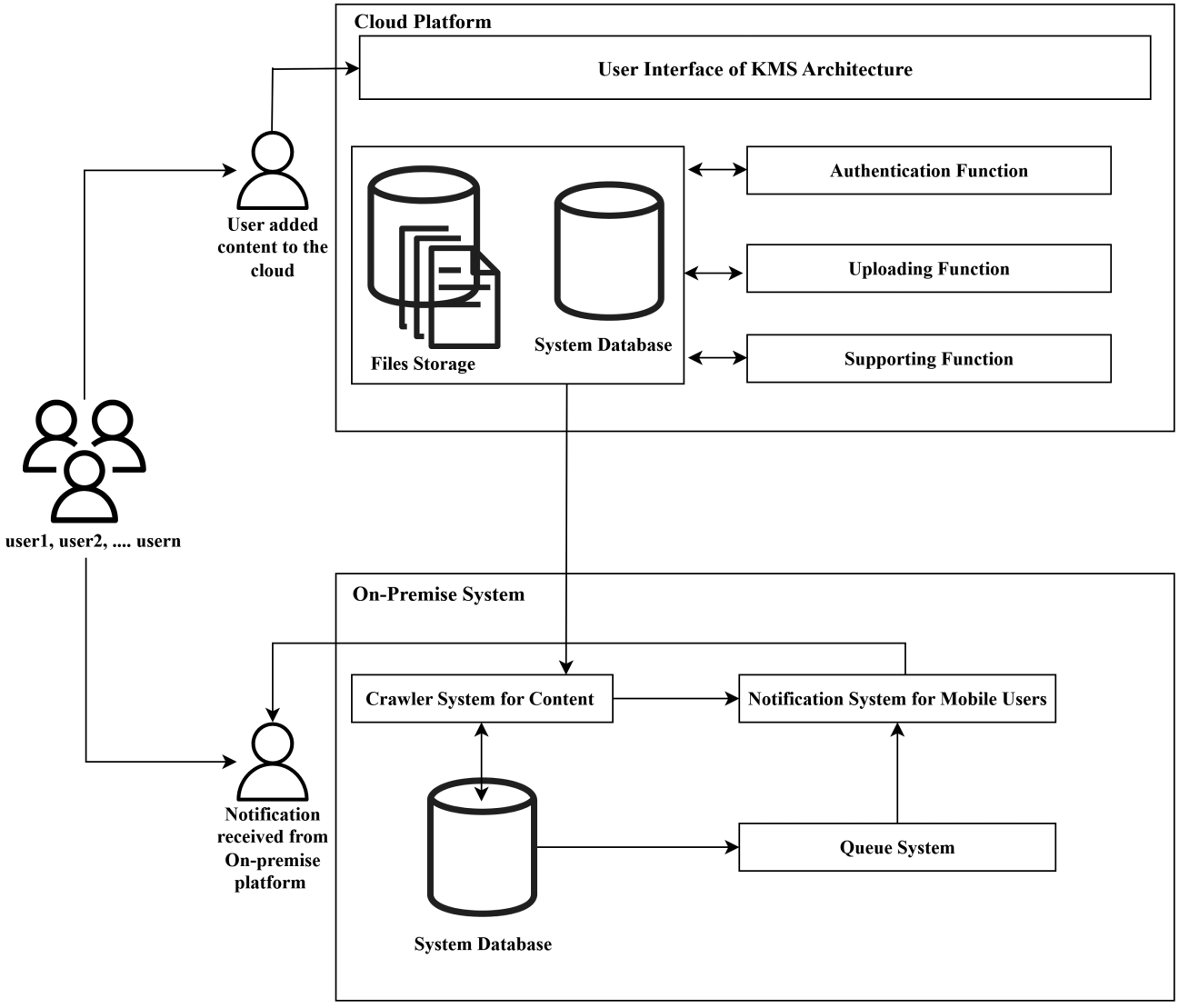


FIGURE 9 | KMS architecture for sharing and receiving the data through the notification.

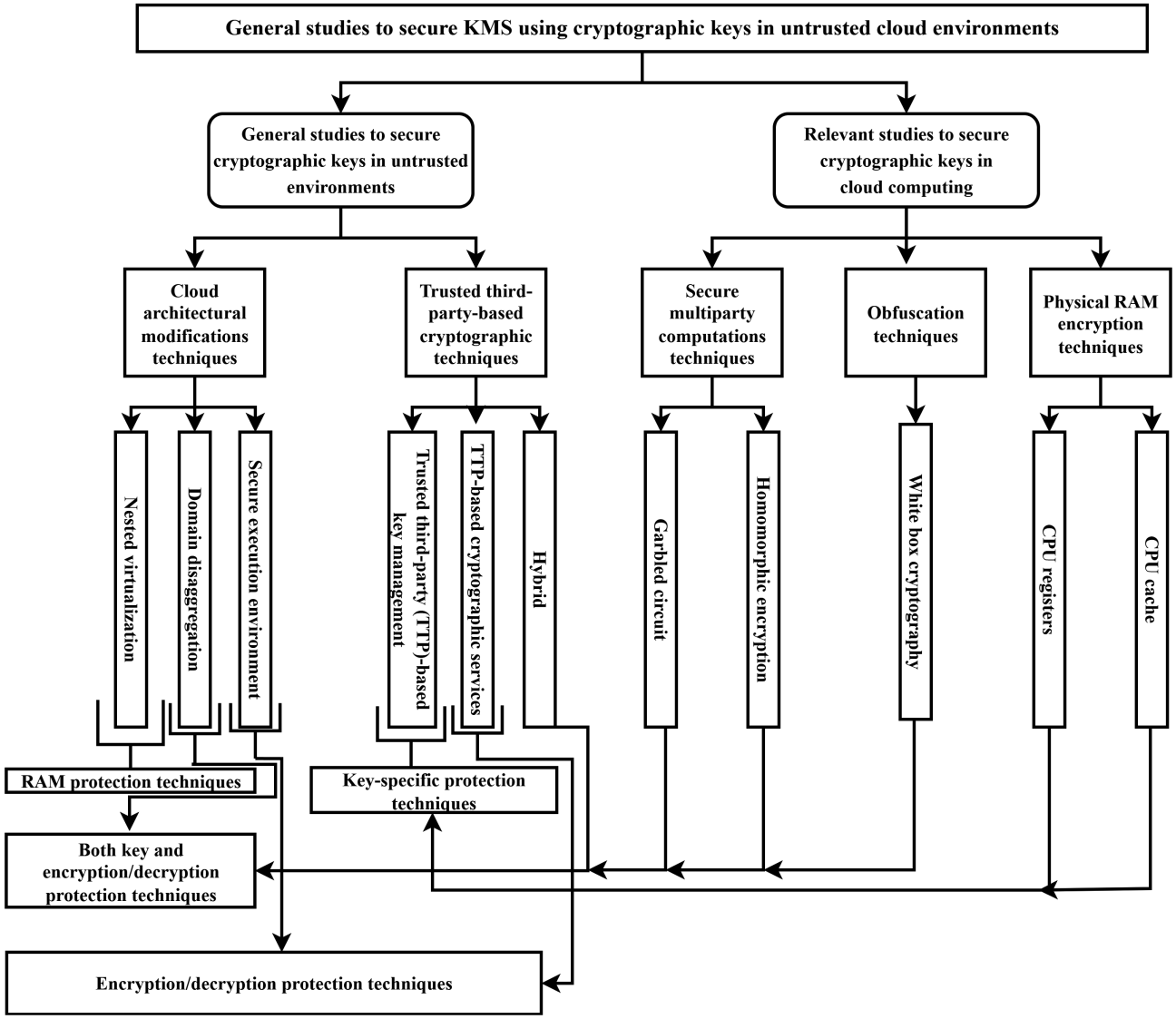


FIGURE 10 | Techniques for securing KMS using cryptographic keys in a CC environment.

Moreover, users who generate content should utilize CC and interact directly with the system. Most content creators are university employees and academics. The Cloud Storage System is the second element [152]. The Cloud Storage System is a vital component of CC infrastructure due to its demanding functionalities and fluctuating resource requirements. Leading cloud service providers like Amazon Web Services, Microsoft Azure, IBM, Oracle, and Google Cloud offer robust cloud storage solutions [153–154]. The User Interface (UI) component (the first point of contact with the user content generator) is a part of this element (as illustrated in Figure 9). This component is crucial for users to add content to the system with the greatest comfort and ease possible. The User Interface (UI) is a critical aspect of the system. It should be visually appealing and intuitive to use. The UI interacts with the Authentication Function, which regulates user permissions and system access. This ensures that users have control over system behavior. The Uploading Function is a central component of the Cloud platform. It communicates with both the authentication function and the supporting function components. The uploading function stores content and associated metadata within the storage system. The supporting functions component includes features like search, sort, filter, summarize, index, and group. The storage media, consisting of file storage and a system database, is the destination for all content. Data is saved as files or database objects within this component [155].

The techniques for securing KMS using different cryptographic keys in CC are shown in Figure 10.

5.5 | Cloud Key Management Services: Problems and Possible Solutions (RQ5)

Cloud subscribers must rigorously protect the public and private keys used to authenticate themselves and sign VM management commands on their system [156]. Although Diffie-Hellman keys are temporary and do not require long-term storage, ensuring secure connections to IaaS cloud services, including both primary and auxiliary services, remains challenging for application users. For secure sessions, a service instance needs an asymmetric key pair and a client key pair, optionally. The IaaS cloud provider's

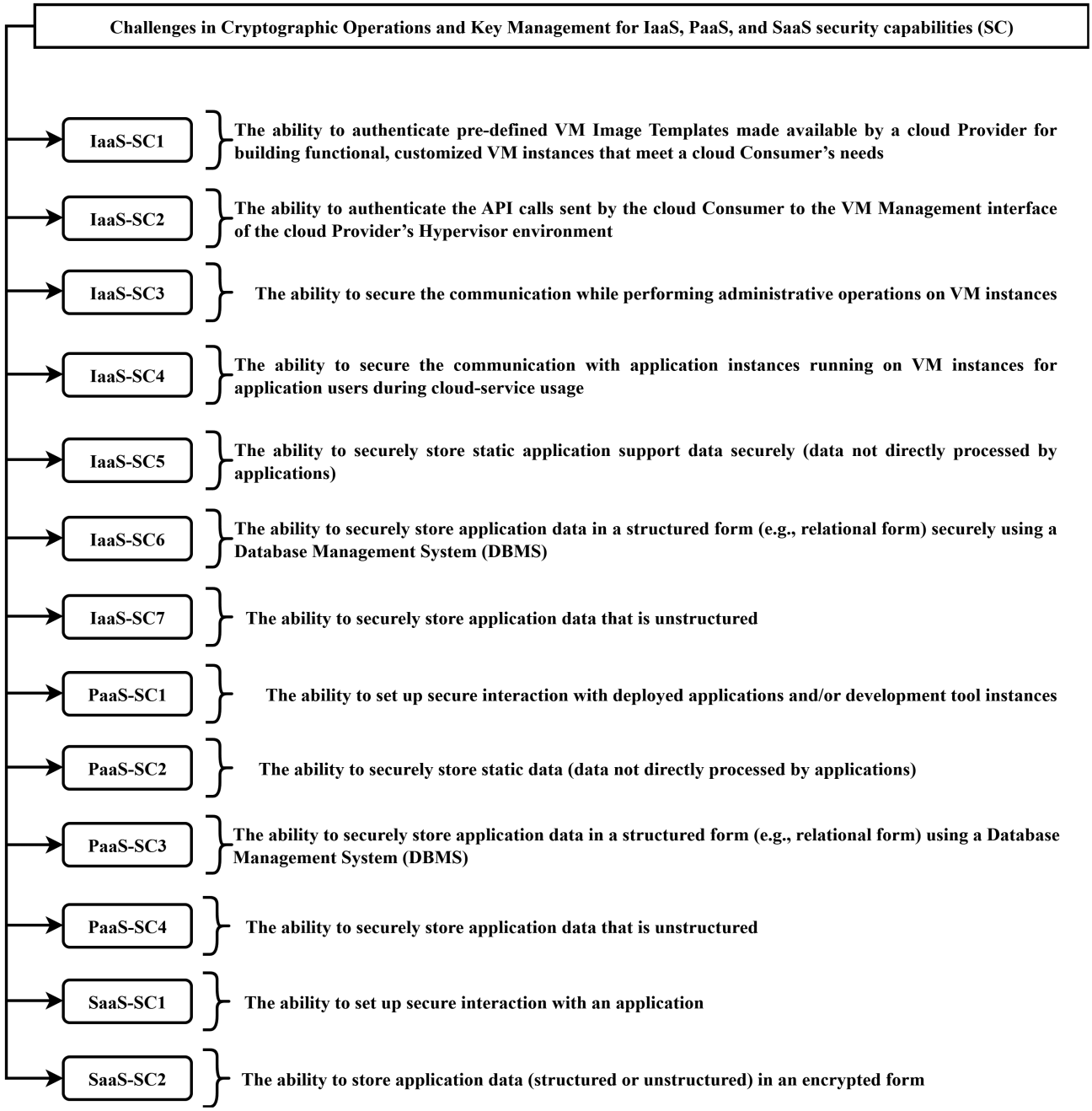


FIGURE 11 | IaaS, PaaS, and SaaS security challenges in cryptographic operations and KMS [159].

KMS should manage the server-side private key, whereas the client-side private key can be controlled by an enterprise KMS. Before a KMS call to retrieve the necessary keys, a mapping between the user's session permissions and the key group must be established using an additional security service [157]. The DBMS instance should ideally operate in a different cloud from the security server, KMS, and permanent key storage to enhance security. The security server and KMS perform role-to-key mapping and key retrieval using the DBMS user's validated credentials. Although logged in, the keys are stored in a cache for the user session. Obtaining the key from the KMS and securely delivering it to the cloud application presents additional challenges. Although a secure connection to the cloud-based DBMS application might seem sufficient, executing the DBMS application, security server, and KMS in the same cloud exposes crucial data to potential risks [158]. Figure 11 illustrates the cryptographic and KMS challenges for IaaS, PaaS, and SaaS based on their security capabilities. Implementing these essential management specifications in a cloud environment over insecure public networks poses significant hurdles [160]. This research analyzes the reference architecture for CC for the three main cloud service types—IaaS, PaaS, and SaaS—identifying core features, necessary security capabilities, architectural solutions, and KM constraints [161].

A PaaS platform aims to provide users with a suitable computing environment and a collection of tools for developing or deploying applications [162]. The consumer is aware of the underlying OS platform that houses the concepts and tools. Still, he or she

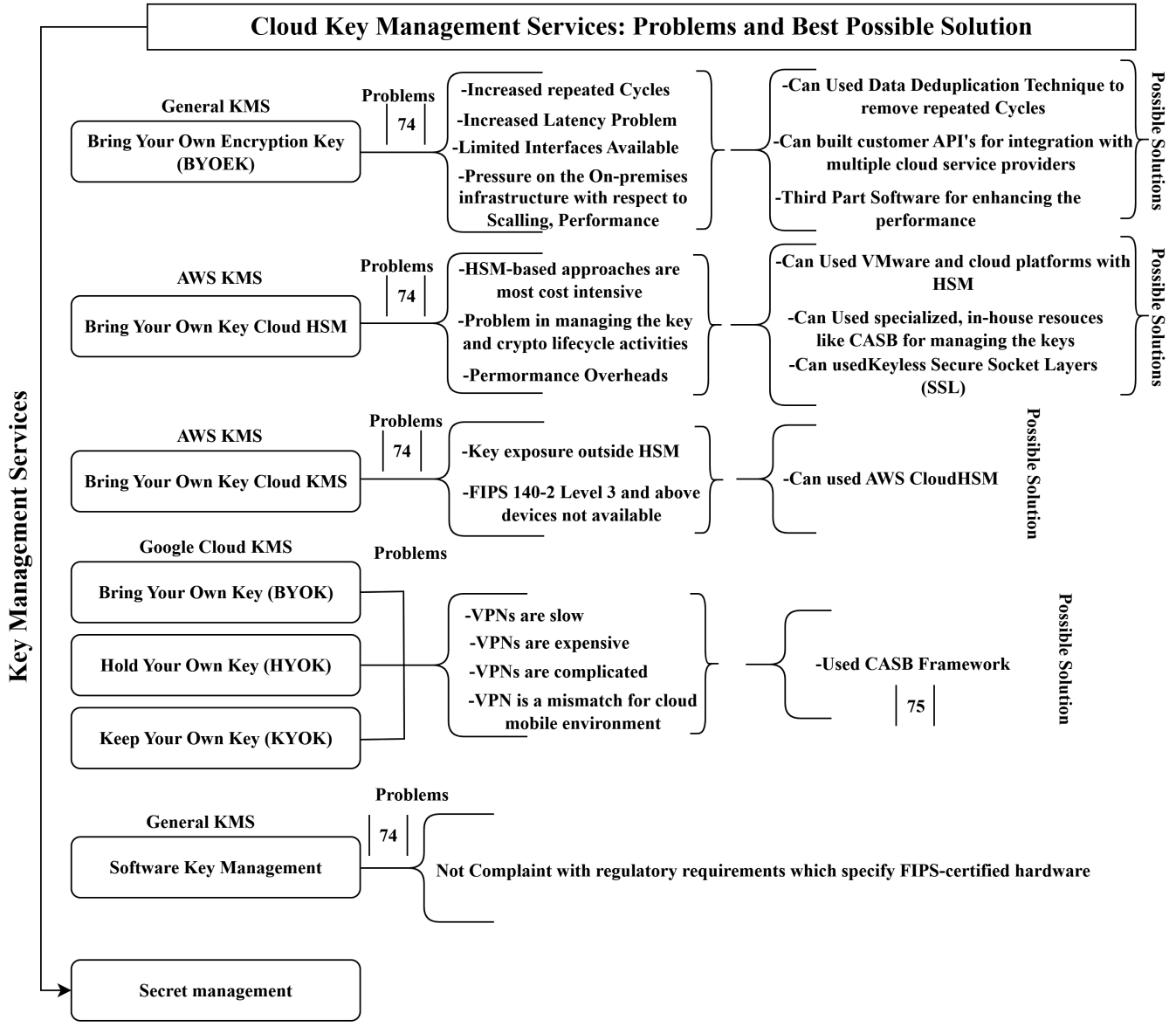


FIGURE 12 | Problems and best possible solutions in cloud KMS [99, 164].

has no control over how it is configured or the operating environment that results from that configuration. The users of these tools interact with them to construct custom apps (and supporting data, such as development libraries). Users might require a storage system to monitor both applications and related data for app performance evaluation.

SaaS platforms provide access to applications that are located on the cloud provider's servers [163]. Therefore, a SaaS cloud user wishes to establish a secure connection with these application instances (via secure transactions and robust authentication) and utilize the various capabilities of the applications as per the permissions given or by adopting the roles allocated to them (which offer the permissions). Furthermore, some SaaS clients opt to maintain data generated or processed by these applications in an encrypted format to (a) safeguard their corporate data from potential cloud provider media loss and (b) prevent unauthorized access by SaaS co-tenants or cloud provider administrators. Figure 12 illustrates the KMS: challenges and their optimal solutions.

Cloud KM generally falls into two categories: entrusting keys to the cloud provider or maintaining them in a customer-owned security infrastructure. Traditional HSM- or TPM-based KM offers high security but is costly and difficult to scale. Bundling HSMs with VMware and cloud platforms like AWS and Azure has made them more accessible [165]. Additionally, the typical multi-user, multi-key cloud situation is not supported by AWS KMS. In [166], three KM services are used, that is, AWS CloudHSM, Keyless Secure Sockets Layer (SSL), and STYX. The security of the key is guaranteed by the services described above, but performance is degraded due to the latency between the content delivery networks (CDN) node and the user. As a result, a solution that satisfies both security and scalability is required.

One of the most important aspects of cloud-based computing is the safe administration of the resources connected to cloud services.

Although cloud services offer widespread computing, flexible resources, and customizable options at lower costs, they still require numerous cryptographic operations for the following reasons.

- Ensure the cloud subscriber's reciprocity with a variety of services.
- The safe storage of data generated or processed by these services.

The KMS required to execute cryptographic operations for roles may be hindered by disparities in ownership and administration of the underlying infrastructures hosting the KMS and protected resources. Although cloud clients technically own content in cloud services, it is stored on devices managed by the cloud provider. Running the KMS needed to manage cryptographic keys, safeguarding this data, often involves utilizing the cloud provider's computational resources. This complicates a cloud

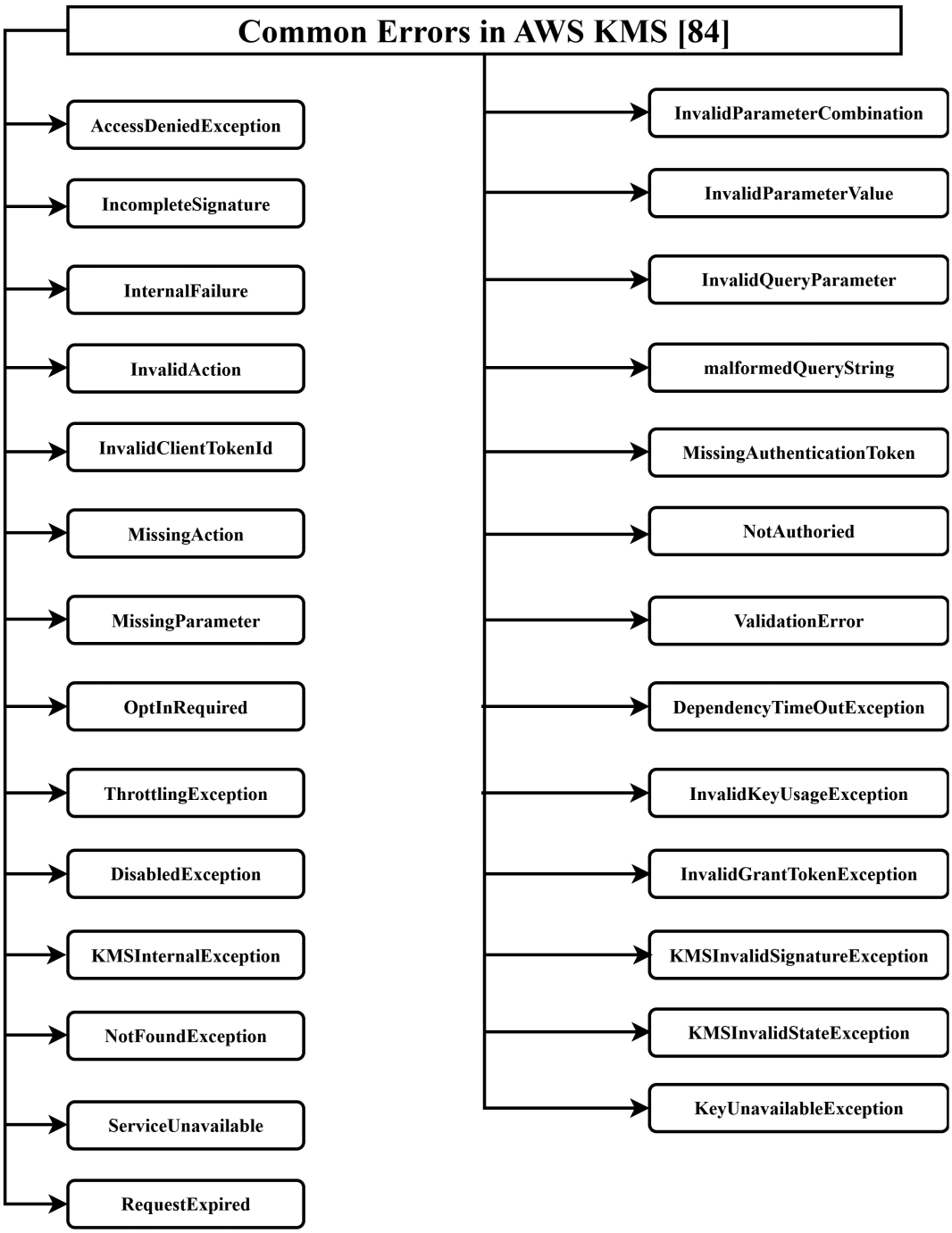


FIGURE 13 | Common errors occur in AWS KMS [167].

TABLE 7 | Availability of major public cloud service providers in the market.

Cloud service provider (CSP)	Number of key managed services	Name of KMS offerings	Overview of KMS offering
Alibaba Cloud	1	Alibaba Cloud KMS	Secure key and cryptography management, custom key rotation, key usage logs, and compliance assistance
AWS	2	AWS KMS, AWS CloudHSM [89]	KM for various cryptographic procedures, hardware security modules, and CloudHSM cluster
Azure	2	Azure Dedicated HSM, Azure Key Vault	Specialized HSMs for cryptographic key storage, certificate and KM, and secret management
Google Cloud	2	GCP Cloud KMS, GCP Cloud EKM	KM is like an on-premises, hierarchical key structure, with a delay for material destruction, and external KM for encryption
IBM	2	IBM Cloud HSM, IBM Key Protect	Secure key storage in tamper-proof hardware, a comprehensive KM solution, and secure key transfer

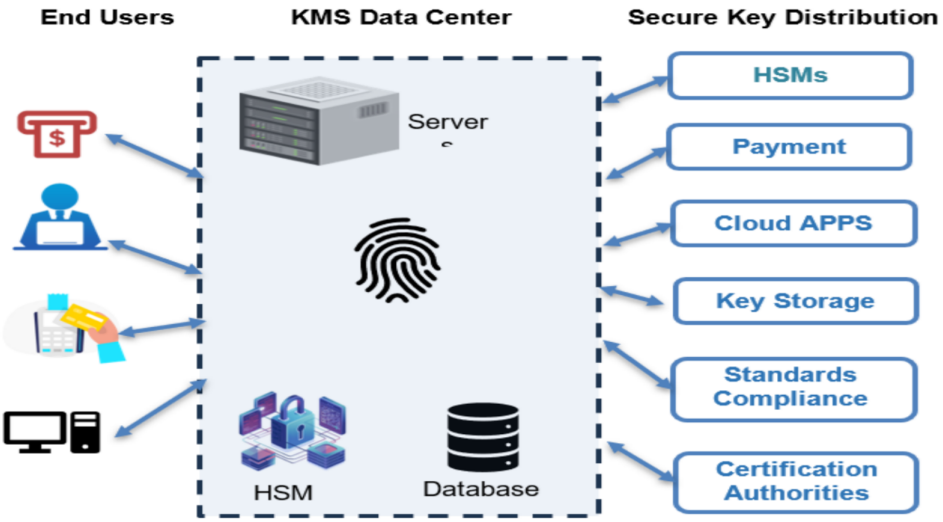


FIGURE 14 | Architecture of KMS with functions and applications.

user's ability to obtain security assurance from cryptographic activity. Figure 13 illustrates common errors in AWS-KMS.

5.5.1 | Big 5 Enterprise-Based KMS Comparison

The KMS offerings of four major public cloud service providers are summarized below [168]. Table 7 provides an overview of KM services offered by the four main cloud service providers: AWS, Microsoft Azure, Google Cloud, and Alibaba Cloud. These services enable businesses to securely manage and protect cryptographic keys used for encrypting sensitive data. Key features include custom key rotation, compliance assistance, hardware security modules, and integration with other cloud services.

5.5.2 | Architecture and Framework of KMS

There are numerous architectural combinations of solutions for CC and KM, as shown in Figure 14. The following are the primary questions that influence how organizations approach the selection of architecture:

1. Which KMS system is the company currently using?
2. What functional, operational, and business results are anticipated from using the cloud service?

Compared to the response to the second question, the first question's answer is less critical. Given that some cloud services do not permit or facilitate integration with an existing KMS, it is probable that the answer to the first question is irrelevant. The organization should adequately document the answer to the second question before tackling the first question. With concerns for cost, complexity, stability, service level agreements, privacy, transparency, regulatory compliance, and adoption speed, the second question may yield a lengthy list of desired results. Factors like interoperability, configurability, standard compliance, and the organization's access to KMS-related expertise may influence

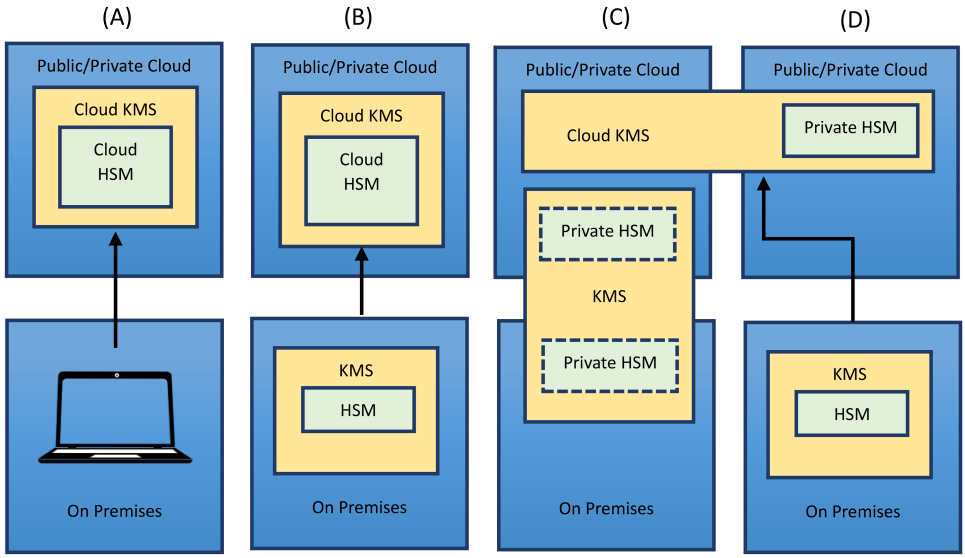


FIGURE 15 | Pattern for cloud KMS.

these outcomes. Given these factors, every architectural design has advantages and disadvantages. It is vital to note that every cloud represented in the architectural patterns illustrated in subsequent sections could be either a public or a private cloud. The architectural patterns can be generalized for implementation or combined with one another to use inter-pattern integration (e.g., a multi-cloud KMS). The architecture of cloud services and the KMS that support or may be integrated with numerous cloud services have seen significant evolution during the past 10 years. With cloud scale and performance, KMS features previously only available as enterprise data center hardware owned by a single company have been made available to all. Due to the previous difficulties with legacy enterprise KMS implementations—cost, complexity, brittleness, and a potential to become "senior citizens" with little desire for revitalization—many businesses will be drawn to cloud KMS choices because of this transition. The capacity to pivot, relocate, or embrace new patterns of cloud KMS is advised for enterprises because the sector is undergoing rapid evolution. It is essential to assess a vendor's capacity to switch to a different pattern or combine patterns when evaluating their implementation of a particular cloud KMS pattern. With more flexibility, businesses will be more agile and will have less friction during mergers, acquisitions, and divestitures.

The patterns seen in Figure 15 for the cloud KMS are as follows:

- (A) is a cloud service that utilizes an internal KMS (including an HSM).
- (B) enlarges pattern (A) to permit the import of important information from a separate KMS.
- (C) is a cloud KMS with a dedicated (private) HSM that is under the control of the owning organization but is physically hosted within the cloud provider's data centre(s).
- (D) shows an example of an on-premises KMS that is used for multi-cloud KMS integration/management and can be hosted either on-premises or in the cloud and is connected to an on-premises cryptographic module such as an HSM or crypto card.

5.5.2.1 | Cloud Native Key Management System (CNKMS). The CNKMS is represented by Figure 16.

The fact that KMS was developed and is owned by the same provider that offers the client's cloud service, and that all KMS's components are in the cloud, makes this cloud architecture pattern distinctive. The KMS may be a stand-alone service provided by the same cloud service provider or a crucial part of the service being used. The cloud provider chooses and controls this setup's cryptographic modules and algorithms.

Attributes:

Except for publicly available documentation, the KMS is often a "black box" to the customer, and all hardware and software components are under the provider's control. Except for publicly available documentation, the KMS is often a "black box" to the customer, and all hardware and software components are under the provider's control.

- Usually, there is little to no separation of responsibility between the KMS features and the cloud service.
- SLA enforcement model with the least disruptions.
- Although this pattern frequently makes it simple to extend KMS functions to other services within the same provider, it may have limited extensibility of KMS functions outside of the provider.
- Pricing is frequently included in the cloud service that the customer purchases and is therefore concealed, difficult to make out, or assumed to be zero.
- Typically, it has great performance and scalability.

Challenges:

This model assumes that different cloud providers have different designs, so each cloud provider will need a different set of technical skills and knowledge. Additionally, the

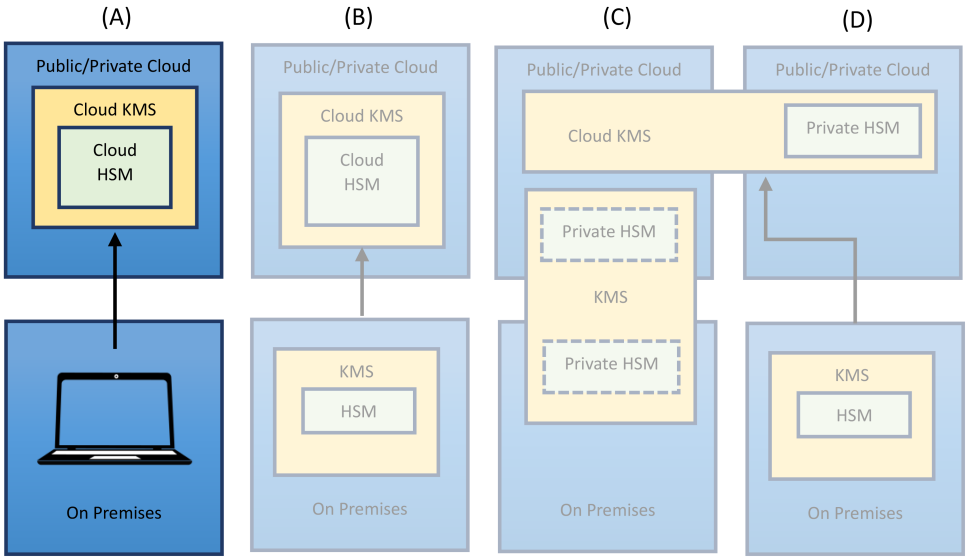


FIGURE 16 | Native pattern for Cloud KMS.

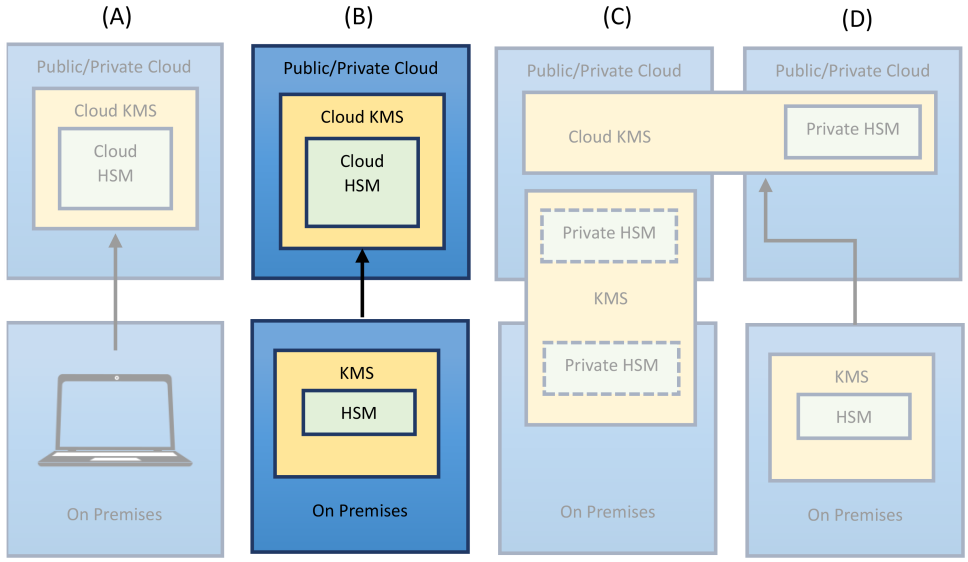


FIGURE 17 | External key origination.

administrative model frequently has a poor separation of roles because it does not distinguish between the KMS administrator and the application/service administrator. However, the consumer-accessible feature set and ease of deployment may lessen the requirement for extensive technical knowledge development and the associated administrative load. Although broad technological characteristics may be specified in cloud provider literature or auditor materials, details regarding specific algorithms and protocols are typically not exposed to the consumer. This architecture might not satisfy all compliance standards because the user often has little to no control over KMS setup or actions, such as key roll (or rotation) period or revocation and recovery. Lastly, because the CSP (Cloud Service Provider) has the keys, this pattern does not completely guarantee the privacy of client information from the cloud provider.

Use Recommendations:

- When are there no apparent reasons for the buyer to use other models?
- When businesses lack KMS experience and/or operational support capabilities?
- When the consumer's cost is driven by variables other than personnel for KMS expertise, such as storage and transactions?
- When is the speed of adoption a crucial consideration?

5.5.2.2 | External Key Origination. The external key origination has been depicted in Figure 17.

Expanding on the Cloud Native paradigm, the External Key Origination pattern enables key creation ceremonies that start with an external KMS. This design is generally established in response to a regulatory mandate that, in some way, targets crucial "ownership," frequently using ambiguous language and focusing on implementation details rather than technical and/or business goals. As there is no interaction between the external KMS and the cloud KMS in this arrangement,

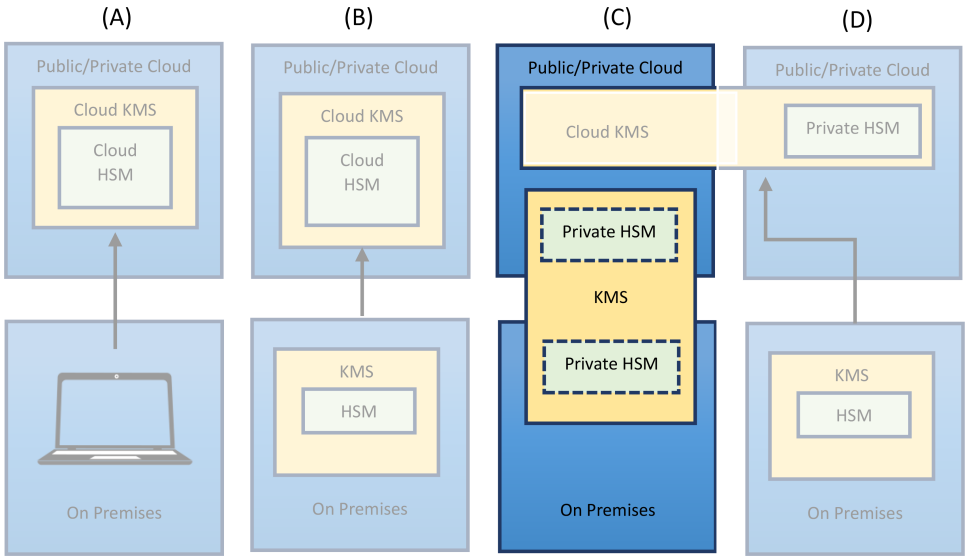


FIGURE 18 | Cloud service using external KMS.

the external KMS merely serves as a key generation facility. Although it has traditionally been discouraged by cloud service providers, in some emerging use cases, the cloud service provider may be required to make calls outbound to the external key manager.

Attributes:

The cloud provider controls the hardware and software components, and the customer has control over a KMS that is external to the cloud and generally uses it for root key generation and import to the cloud KMS.

- The procedure of importing external keys may impact recovery operations, but the possible impact on cloud SLA is minimal.
- It provides a clear separation of roles for root key generation.
- This pattern may have low extensibility of the KMS functions outside the provider but frequently allows easy extension to other services within the same provider.
- The licensing of an external key import feature typically determines pricing.
- Cost is typically determined by licensing the key import functionality, the number of keys imported, and managing the external KMS from which key imports are made.

Challenges:

All the challenges of the Cloud Native pattern apply here, and additionally, the cloud service provider might only support a few hardware producers for external key creation. As a result, acquiring suitable KMS hardware and software would be needed, which might require learning new skills and extending the implementation process.

Use Recommendations:

- When does a customer require additional External KM key custody via approved emerging models?
- When does a consumer require a specific key ceremony that cannot be satisfied by the Cloud Native pattern?

5.5.2.3 | Cloud Service Using External Key Management System. When using an external key management system (EKMS) (As shown by Figure 18), the KMS is completely hosted outside of the cloud service, either entirely on the client's premises, wholly hosted by a third party that the client selects, or a combination of the two. Hardware may belong to the customer or the cloud provider, but it is only provided for the customer's use. In a co-location paradigm, where consumer hardware is housed in the same building as the cloud provider, a cloud provider may enable a dedicated cloud HSM service offering. If the KMS is the cause of a service incident, the consumer typically consents to service-level agreement considerations and manages all parts of the KMS.

Attributes:

- With significant customer configuration and control, this pattern offers the greatest portability because most or all KMS can be combined into a single point of management and can support complete separation of duties for KMS and cloud service activities, as well as within the KMS.
- Can ensure that the key material is never shared with a cloud service provider, ensuring that no consumer plain-text data is exposed to the provider.
- Can meet customer key creation ceremony requirements with customer-driven FIPS compliance for all keys.

Challenges:

Since this model is less common than other cloud architectures, it might not be offered for the service selected. Using this paradigm has implications for service-level agreements, and the customer would not be able to get compensation in the event of service failures. This model is typically incompatible with SaaS or any service where the cloud provider's systems need to process

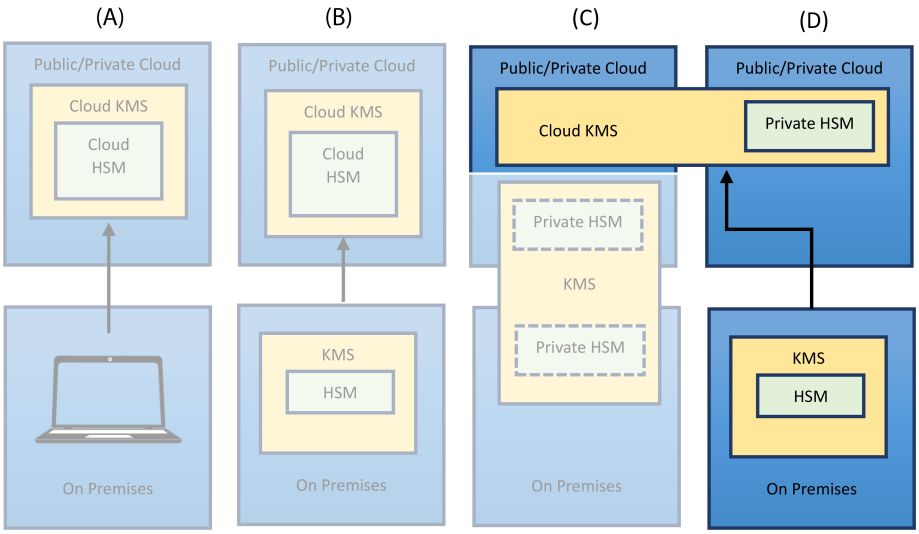


FIGURE 19 | MCKMS pattern.

customer data in plain text. The general justification for wanting to utilize it is to retain the complete secrecy of the client's plain text. To completely manage the KMS, the customer must possess the necessary capabilities.

Use Recommendations:

- When a consumer wants to utilize the same KMS for both on-premises and cloud services?
- Can it enable the potential to obtain a higher FIPS certification level?
- Can it support the necessity of keeping plain text utterly confidential from the cloud provider?

5.5.2.4 | Multi-Cloud Key Management Systems (MCKMS). The Multi-Cloud KMS pattern (as shown in Figure 19) demonstrates how different KMS implementation strategies can be combined with cloud services. Existing cloud services that accept an external KMS allow it to span multiple KMS options for the cloud.

Attributes:

- The other cloud KMS patterns deployed affect this pattern's characteristics, including cost, complexity, control, implementation time, scale, performance, and compatibility.
- The ability to use one cloud KMS as a backup to another cloud KMS enables cloud-scale fault tolerance, and earlier investments in knowledge and resources may have the unintended side effect of accelerating the adoption of new cloud services.

Challenges:

This framework is often reserved for a KM strategy. This is because it requires a broad amount of knowledge, takes the longest time to implement, and costs the most in capital or operational costs, or even both. Despite this, it offers the highest level of portability, fault tolerance, and scalability because it uses several different public cloud providers.

Use Recommendations:

- Organizations with dynamic composition frequently engage in acquisitions and divestitures.
- Complex compliance requirements and situations where resilience and scalability are highly valued [169].

5.5.2.5 | Real-World Case Studies of KMS Implementation. Real-world implementations of KMS in CC highlight both successes and challenges:

- *AWS Key Management Service (AWS KMS):* AWS KMS, integrated with services like S3 and EBS, exemplifies the Cloud Native pattern. A 2023 case study by Susanto et al. demonstrated its use in a financial institution, achieving seamless key rotation but facing latency issues during peak loads, underscoring scalability challenges (Susanto et al. 2023).
- *Google Cloud KMS with External Key Origination:* A healthcare provider adopted Google Cloud KMS with an external HSM for HIPAA compliance, importing keys from an on-premises system. This reduced provider access to plaintext data but increased operational complexity (Google Cloud 2022).
- *Microsoft Azure Key Vault in Multi-Cloud:* A multinational retailer implemented Azure Key Vault across Azure and AWS, leveraging the MCKMS pattern. Tang et al. (2024) noted improved fault tolerance but highlighted integration costs as a persistent challenge.

The real-world case studies to strengthen the practical angle are given in Table 8.

Table 8 demonstrates how Netflix and PayPal employ cloud services for their data security, including KMS by AWS and Google Cloud. Netflix and Airbnb apply AWS KMS for their centralized key management and their integration with the cloud, which, along with their compliance support, offers them a lot in terms of workflow. In the same manner, Snap Inc. and PayPal utilize Google Cloud KMS, with PayPal applying EKM for better control

TABLE 8 | Real-world case studies to strengthen the practical angle.

Cloud provider	Organization/ use case	Context	Benefits observed	Challenges faced
AWS KMS	Netflix	Netflix uses AWS KMS to protect streaming content and user credentials by managing encryption keys for Amazon S3, DynamoDB, and custom applications	Centralized key management, integrated with AWS services, supports compliance (e.g., PCI-DSS, GDPR)	Key usage quotas in high-volume workloads require architectural planning
AWS KMS	Airbnb	Airbnb secures sensitive user data (e.g., ID verification, payment) using AWS KMS with envelope encryption in microservices	Seamless integration with Lambda and ECS; provides an audit trail using AWS CloudTrail	Latency concerns with frequent KMS calls; partially mitigated via local caching
Google Cloud KMS	Snap Inc. (Snapchat)	Snapchat uses Google Cloud KMS for encrypting messages and media stored in Google Cloud Storage	Scalable encryption services with regional key storage options; meets data sovereignty needs	Complexity in managing the lifecycle and rotation of customer-managed encryption keys (CMEK)
Google Cloud KMS	PayPal	PayPal uses Google Cloud External Key Manager (EKM) for retaining complete control over cryptographic keys while using GCP infrastructure	Enhances trust in cloud security; complies with strict financial regulations	High integration overhead; requires custom hardware security modules (HSM)

over their keys. Google Cloud KMS has its advantages of scalability and data sovereignty, but Snap Inc. struggles with the freedom of having complex key lifecycle management. Furthermore, Snap Inc. runs into the problems of administering automated control through these frameworks, whereas PayPal's implementation of EKM leads to high effort for tightly coupled components and custom-built devices.

5.5.3 | Toolkits to Design and Develop Functionalities of KMS

The procedures that control the production, usage, storage, and destruction of keys are known as the KM life cycle/phases. The main actions involved include creating, enabling, sharing, disabling, breaching, pausing, ending, storing, restoring, renewing, canceling, substituting, and eliminating. The state of the key has been depicted in Figure 20. A detailed description of the same has been given below.

Generation:

Although the key has been generated, it has not yet been approved for usage. Therefore, the key can only be used to confirm or prove possession in this condition. Symmetric keys or public/private key pairs should be generated only when necessary.

Activation:

The private key can be used to encrypt or decrypt data and to create or verify digital signatures. These keys should only be activated when necessary to avoid security risks.

Distribution:

After the key generation, key material like Random Bit Generators (RBG) must be transferred between more entities. This process of key sharing and key material transportation can be done manually or automatically.

Deactivation: Deactivated keys can still process information that has been cryptographically secured, but cannot be used to apply cryptographic protection. If symmetric or private keys are no longer needed to apply cryptographic protection to data, they should be deactivated. These keys may be archived or destroyed after being deactivated. A public key can expire, be suspended, or be revoked; it cannot be made inactive.

Compromise:

Generally, keys are often leaked or chosen by people who are not supposed to have them. So, data should not be protected with a key that someone else knows. But sometimes, we can still use encrypted data even if the key is bad or if the public key matches a bad private key.

Revocation:

If the key has been revoked, it can still be utilized for processing (i.e., for scenarios other than a compromise). When using X.509 certificates, information about revoked public keys is sent to those who rely on these certificates through CRLs or OCSP responses. When secret keys, including private keys, have been compromised or discovered in CRLs or OCSP responses, they should be revoked. After that, the key should be added to a compromised key list.

Suspension:

A suspended key should not be utilized to enforce cryptographic security (e.g., encryption of plaintext or generation of a digital signature). However, a key that has been suspended may still be used to process data that has been protected before the suspension (e.g., decrypting ciphertext or verification of a digital signature). In addition, a key or key pair that has been suspended may later be activated again, deactivated, deleted, or

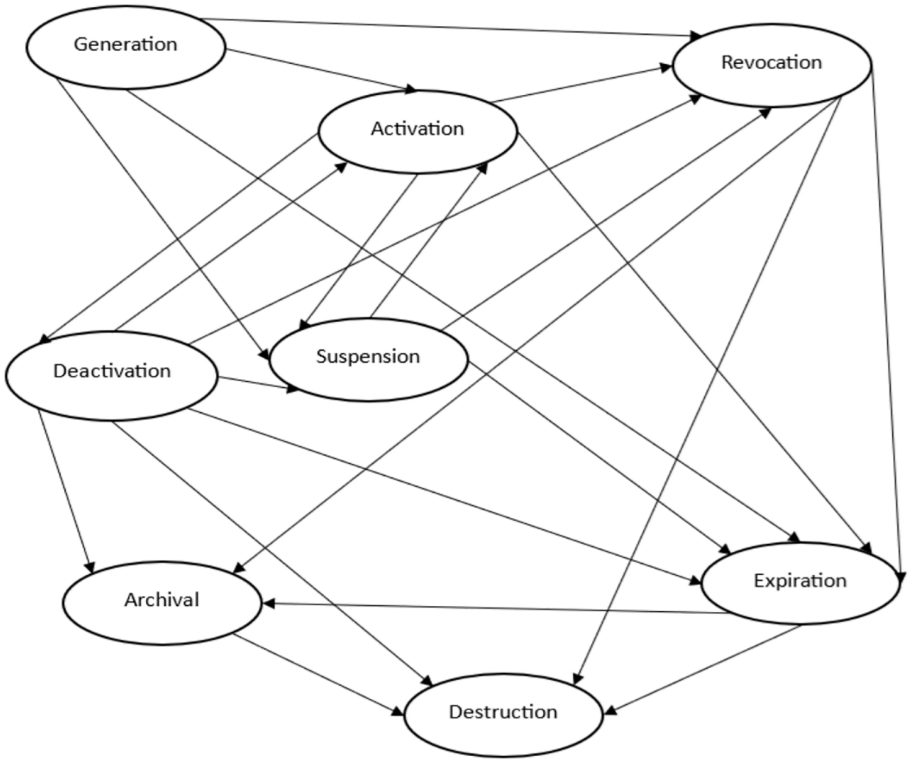


FIGURE 20 | The state of the key.

moved to the compromised state. Keys may be temporarily deactivated if their owners are absent or there is uncertainty about who can use them. The appropriate parties are notified that the companion private key has already been terminated in such a situation.

Rotation:

Key rotation is discarding an encryption key and replacing it with a fresh key. The quantity of data encrypted with each key is reduced during key rotation, reducing the volume of information that would be exposed if a key were stolen. Rotation and crypto-periods are interrelated notions since a proper rotation depends on crypto-periods, considering the volume of data a key is expected to encrypt.

Replacement:

When a key is compromised or its crypto period expires, it must be reset. Replacement can be done using either a key-update method or a rekeying procedure. Rekeying is the process of replacing a key with a new one that is entirely independent of the value of the old key (i.e., knowing the value of the old key does not reveal the value of the key that is replaced, and vice versa).

Expiration:

The crypto period is said to have expired when the authorized usage time of the key ends.

Archival:

Key archival is keeping a certificate's private key so it can be recovered in the future if necessary. When the key is not often used but may still be necessary after the key's crypto period, it is retained. The potential decryption of archival data serves as an illustration for secret or private keys. Verifying previously signed documents that have been archived is an example of public keys.

Recovery:

Restoring cryptographic keys or archived keys to their initial condition is known as key recovery. To maintain the allocated usage attributes, it is crucial to precisely reproduce key attributes that are connected to their original condition. Then, when a key is still in use for encryption, the information can be recovered from backups or archives if it has been archived.

Destruction:

As soon as it is explicitly determined that a key is no longer needed, all the symmetric key copies should be deleted. The keys should be discarded so that no trace of the keying material remains, making it impossible to recover it physically or electronically. Public keys can be kept or deleted as required [168].

5.5.4 | Classification of KMS

The two basic categories of KM strategies are static and dynamic. As depicted in Figure 21 and detailed in the subsequent sections, these classifications are further categorized into several subcategories [170].

5.5.4.1 | Dynamic Key Management (DKM) Schemes. In DKM, each session is assigned a unique key. This means that keys are constantly changing, ensuring that even if one

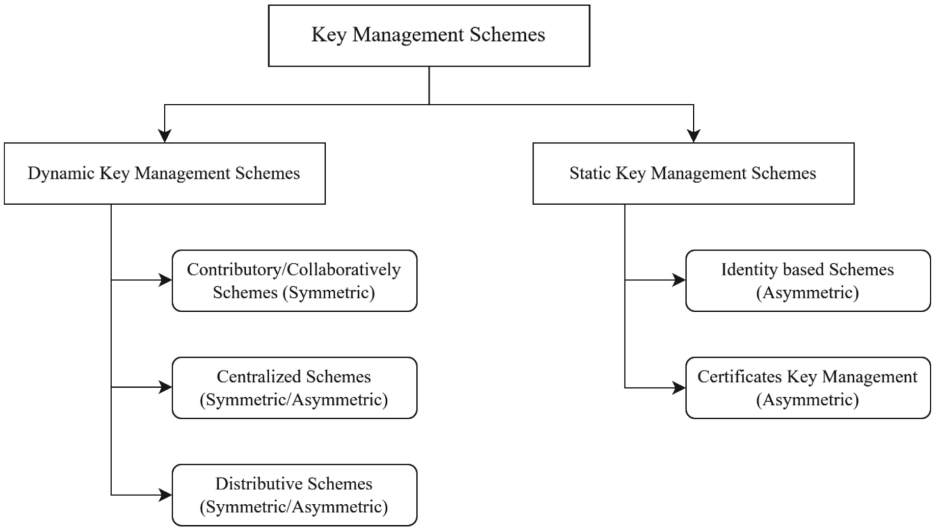


FIGURE 21 | Classification of KMS.

key is compromised, it will not affect future communications. When a session ends, new keys are automatically generated and distributed to the nodes involved without requiring any manual revocation or updates. Since there are three main ways that communication between the sender and recipient might begin—contributory, centralized, and distributive—it is noted that the keys in dynamic KM schemes are established dynamically. Each of these ways will be discussed individually with the support of an example in the upcoming sections. On the contrary, different cryptographic keys are allocated for various sessions under dynamic KM schemes.

- Contributory/Collaborative Key Management (C/CKM) Scheme
  C/CKM methods are symmetric cryptographic-based approaches that are distinct due to the lack of a trustworthy third party, which is usually in charge of creating and disseminating the cryptographic keys. A secret symmetric key must be determined or selected by the agreement of all participating organizations. For a particular current session, the keys can be created in pairs or more for two or more parties. Therefore, it is identified as a session key at times.

- Centralized Key Management (CKM) Schemes
  CKM systems use a trusted authority to create and distribute unique session keys to all group members in the cloud. A semi- or entirely distributed certifying authority verifies the user's public key in these approaches. Any public key user must have the relevant certification authority to validate its authenticity before utilizing it.

- Distributed Key Management (DKM) Schemes
  Distributive schemes do not explicitly have a centralized design because they use one or more authorized entities to distribute keys. These algorithms can make use of both symmetric and asymmetric cryptographic systems.

- Dynamic Key Management Schemes in Zero Trust Security Architecture
  The "never trust, always verify" concept is the main idea behind "Zero Trust Security" (ZTS) architecture. This principle is followed regardless of whether network access requests come from inside or outside the perimeter of the network. In ZTS, KMS rely on strong identity checks, encryption, and fine-grained access control, making key management fundamental in this context. ZTS is enabled through KMS, which is a critical components that generate, stores, and rotates authentication keys for devices, users, and services. These keys are accessible only based on a remote-controlled, policy, role, and contextual (device posture or location) factors framework due to integration with identity and access management (IAM) systems. KMS has also been placed under greater demand due to two additional ZTS principles: just-in-time (JIT) access with continuous verification, devoid of pre-issued credentials. These two principles call for high-frequency key usage, low latency, and auditability. Current solutions to these demands include automated policy-driven key issuance, vertical control through secure enclaves and HSMs, which align with ZTS principles. Overall, these advancements enhance the taut coupling of access decisions accompanied by cryptographic guarantees, thereby strengthening cloud security posture.

5.5.4.2 | Static Key Management (SKM) Schemes. SKM assigns keys to nodes for their entire lifespan. Unlike dynamic KM, which assigns keys for each session, static keys are established through mutual agreement, symmetric cryptography, or centralized certification. Once created, a static key remains valid until it is revoked or the certification changes.

a. Identity-Based Key Management (IKM) Schemes
   A trusted third party, called a private key generator, creates the private key in these systems. The public key is derived from a node's identifier, like an email, IP address, or MAC address. However, these methods have limitations, including a lack of privacy and anonymity due to public keys serving as central identifiers. Public key cryptography is always used to generate key pairs [171] to address

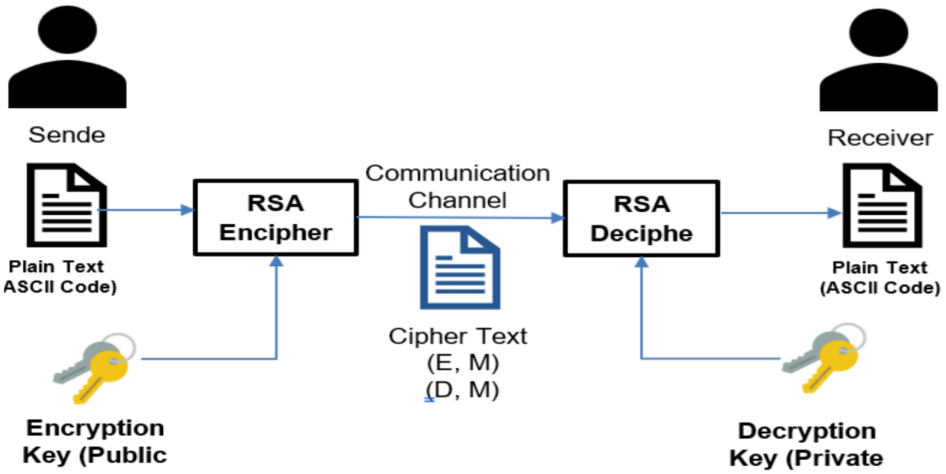


FIGURE 22 | Working of the RSA algorithm.

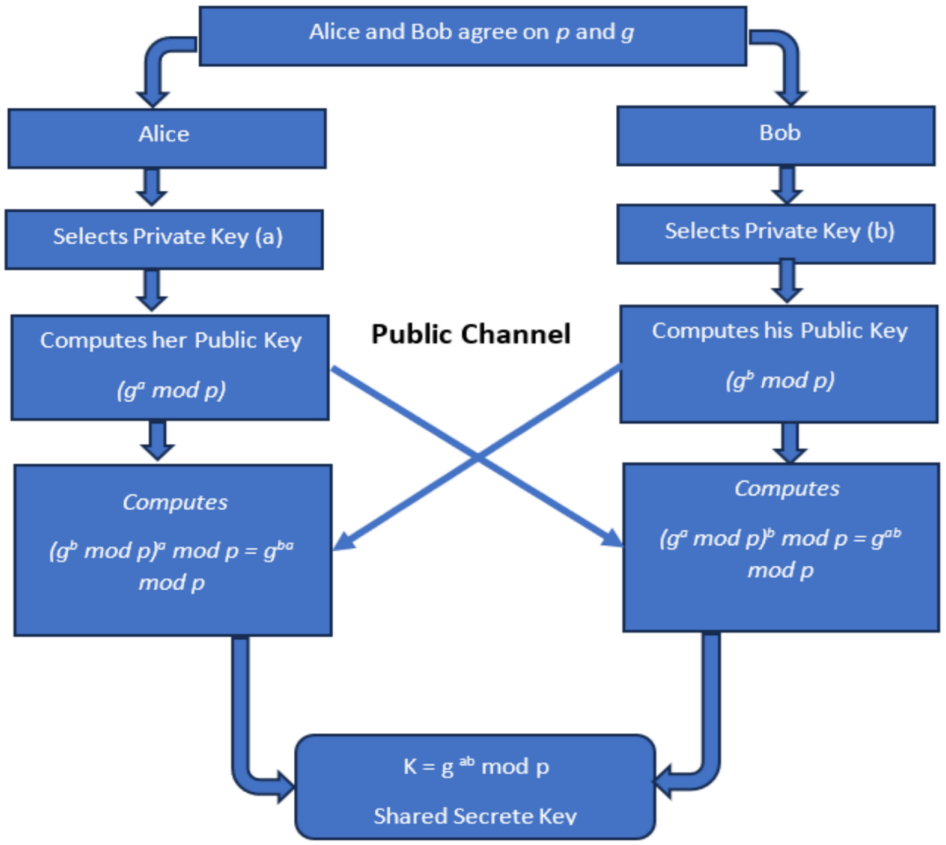


FIGURE 23 | Block diagram of Diffie Hallman.

these issues. The public key, which includes the node's IP, MAC, and email address, is used to create the corresponding private key.

b. Certificateless Key Management (CKM) Schemes
Certificateless KM schemes use certificateless public key cryptography (CLPKC), whereas threshold cryptography systems are used to set constraints and limitations on key generation. These methods act as a bridge between PKI and ID-PKI. Hence, they are exempt from the key escrow issue because they do not require certificates.

The encryption key gets deposited in escrow (a contractual arrangement) in a fair cryptosystem for checks and balances.

5.5.5 | Algorithms of KMS

This section discusses the many cryptographic approaches that are used frequently. There are two sorts of cryptographic algorithms, depending on whether they are symmetric or asymmetric. Asymmetric algorithms are also called public key algorithms [169].

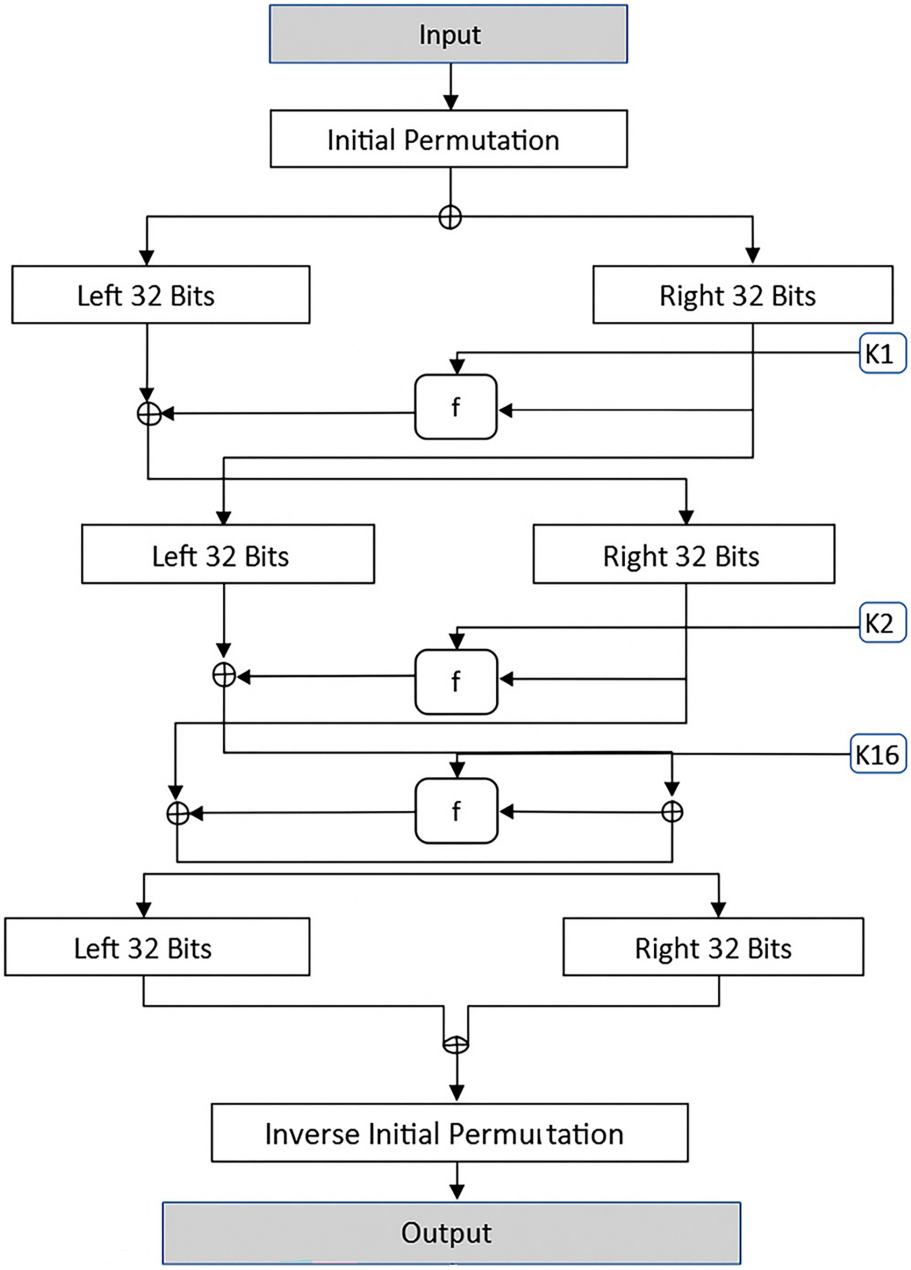


FIGURE 24 | Block diagram of DES.

Asymmetric algorithms include the following:

1. RSA: The three people who proposed the RSA algorithm are Ron Rivest, Adi Shamir, and Leonard Adleman (as shown in Figure 22). The encoding algorithm is a popular public key encoding one. The RSA algorithm employs two keys: public and private keys. Using an openly available public key, the communication is encrypted. A private key is used for message decryption. It is preserved as a secret.

2. Diffie-Hellman: It is a popular method for securely establishing a shared secret key between two parties over an insecure channel. This technique allows parties to exchange secret keys without requiring any prior shared secrets. Unlike symmetric encryption algorithms, which use the same key for both encryption and decryption, Diffie-Hellman focuses solely on key exchange. Before sending a message, both parties must verify the exchanged key to ensure its authenticity. Figure 23 illustrates the Diffie-Hellman algorithm.

Symmetric encryption algorithms can be divided into two main types: block and stream. Some of the most common examples of symmetric algorithms include:

1. DES: The Feistel block cipher is the basis for the Data Encryption Standard (DES) (as shown in Figure 24). This block-based system was designed and developed by a cryptography researcher at IBM.

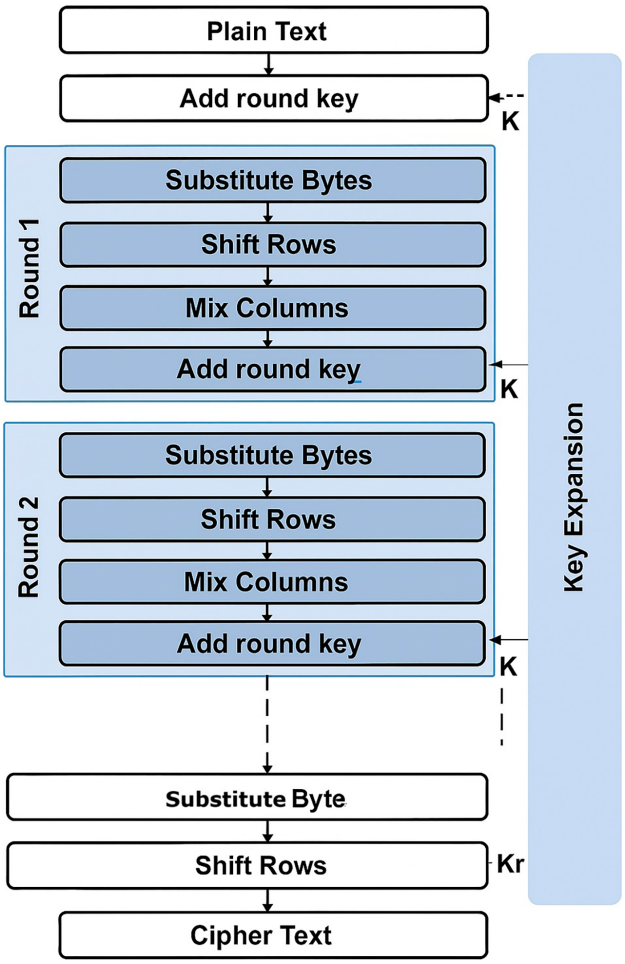


FIGURE 25 | Block diagram of AES.

2. AES: The Advanced Encryption Algorithm (AES) (as shown by Figure 25) was developed to fill in the gaps and flaws in DES, such as the delay in executing it on board. DES is a hardware-based encryption algorithm that is slower than AES, a software-based algorithm. AES can use a 256-bit, 192-bit, or 128-bit key, whereas DES only uses a 56-bit key.

3. Blowfish: The Blowfish algorithm was invented in 1993 (as shown by Figure 26). This encryption method makes use of 38 to 448-bit key lengths. The Blowfish algorithm is a 16-round Feistel cipher that utilizes large S-boxes derived from the encryption key. A 2009 study by S. Abdul Elminaam et al. found that Blowfish was more energy efficient than other encryption algorithms.

5.5.6 | Comparative Analysis of Various Symmetric Encryption Key Algorithms

Table 9 features a synchronized encryption key algorithm comparison of DES, 3DES, AES, Blowfish, HiSea, Speck, ChaCha, and SIMON, which details the year of invention, designers, structural properties, key and block lengths, data processing speed, security levels, and encryption speeds, as well as hardware effectiveness for each algorithm while describing their known vulnerabilities. Symmetric encryption has evolved since DES and 3DES began. It exhibited speed limitations while experiencing vulnerabilities until the present AES, ChaCha, and SIMON, which improve security standards and enhance encryption rates, as demonstrated in the table analysis. Additionally, the table demonstrates trade-offs between security and speed and efficiency when using hardware or software implementations while detailing their resistance against cryptographic attacks.

5.6 | Summary

KM is the process of handling cryptographic keys throughout their lifecycle. This includes generating, distributing, storing, using, validating, and replacing keys. KM involves designing protocols for key servers, user procedures, and other necessary components. It is critical to a cryptosystem's security, ensuring that keys are protected and used correctly. KM is particularly challenging because it involves complex interactions between system policies, user training, organizational procedures, and human behavior. Coordinating these elements is essential for effective KM. These issues go beyond just cryptography engineering. Since KM includes organizational and technological decisions, some KM components risk being overlooked by engineers and administrators who are worried that the issue is either managerial or technical. However, security in the cloud presents unique difficulties that are not present in traditional network or computer security because of intrinsic resource and computing limitations. The potential for study in cloud security is now considerable. In matters of cryptographic security, the key is crucial.

5.6.1 | Implications of Hybrid Cloud Adoption on Federated Key Management

The adoption of hybrid clouds brings about new possibilities as well as challenges for federated key management. Hybrid models combine the private and public cloud infrastructures of an organization, allowing optimization in cost, performance, and data locality. But this diversity also results in fragmented key management for various systems, which increases the complexity of cross-system interoperability, multi-instance scalability, and policy enforcement. In this arrangement, remote key management federations or decentralized key management systems must share or synchronize the keys securely with consistent access control and auditability governance. Azure, AWS, and Google's hybrid configurations increase worries around trust boundaries because different cloud providers use and enforce proprietary, regulated security policies, which makes unified key lifecycle management difficult. In addition, cross-system controlled latency, policy compliance, and secure transfer communication become harder to manage in these distributed systems. Hence, the adoption of hybrid clouds requires the design of federated KMS frameworks that are policy-compliant, scalable, and interoperable, along with standardized protocols and self-reconfiguring trust relationships, fault-tolerant security architectures, and robust cryptography to ensure comprehensive security in multi-cloud environments.

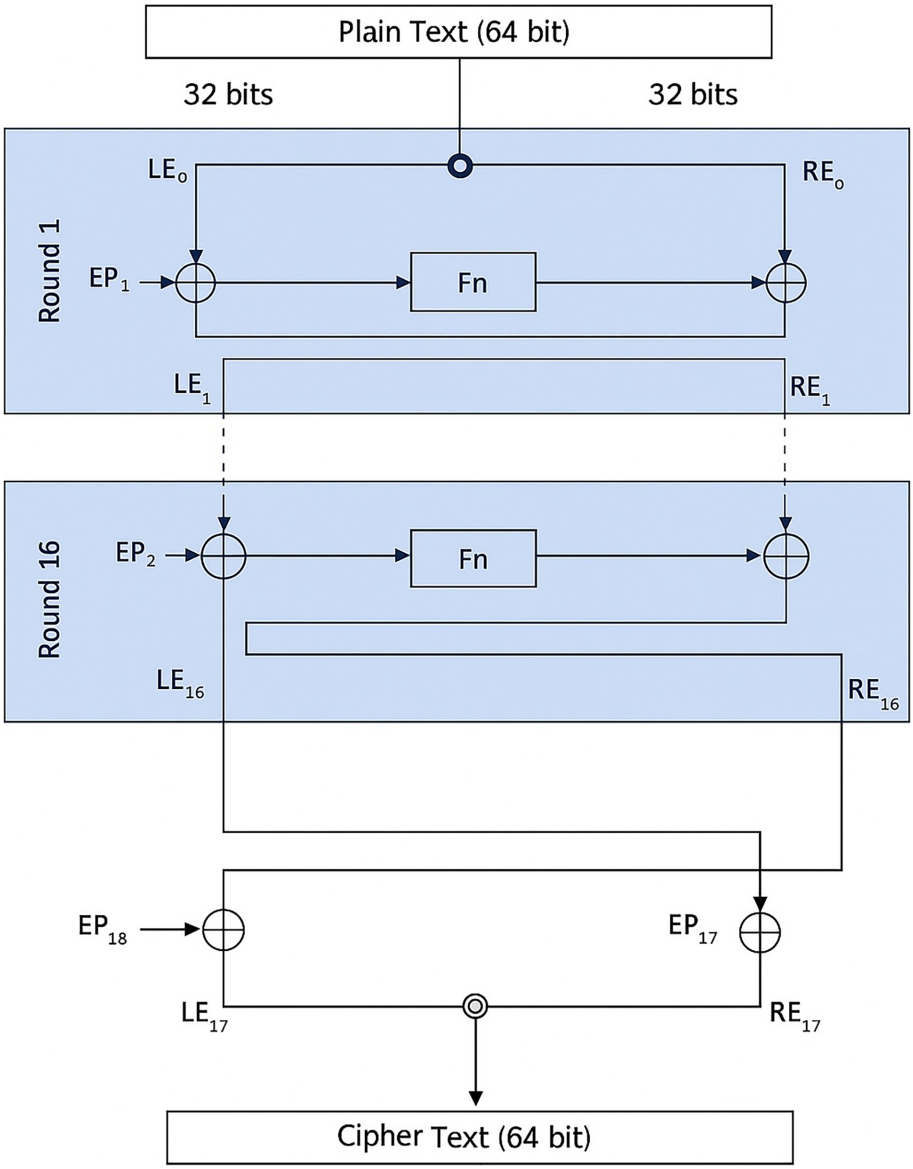


FIGURE 26 | Block diagram of Blowfish.

6 | The State-Of-The-Art: Discussion and Characterization

6.1 | Framework of Characterization

This paper summarizes the key findings of some of the most widely used KM and encryption techniques in the field of cryptography. Firstly, we identify and analyze as many of their flaws and vulnerabilities as possible. Secondly, we explore and suggest the main aspects of these weaknesses that can be avoided or minimized by the correct implementation. Unfortunately, the number of vulnerabilities in the cloud is steadily rising. A powerful encryption method and KMS are needed at hours 151,152,178 to address these issues. This survey paper provides an elaborate and detailed overview of various KM schemes, their uses, encryption algorithms, and a literature survey of both.

6.2 | Basic Observation

KM, data loss, data leakage, and user identity authentication are the primary security issues in CC. Cloud data security is becoming increasingly crucial as more enterprises adopt the cloud model. Since the data is encrypted using cryptographic keys, KM is the most vital factor in convincing potential cloud subscribers and business managers to store their critical data securely in the cloud.

6.3 | Captious Thought on the State-Of-The-Art

This research paper examines current methods used for KM, highlights their weaknesses, and proposes innovative solutions to address these shortcomings. Additionally, it illustrates a com-

TABLE 9 | A side-by-side look at different symmetric key encryption algorithms [172].

Algorithms/ schemes	DES	3DES	AES	Blowfish	HiSea	Speck	ChaCha	SIMON
Year of Invention	1977	1978	2001	1993	2011	2013	2008	2013
Designed and established by	IBM	IBM	Joan Daemen and Vincent Rijmen	Bruce Schneier	Sapiee Jamel	NSA	Daniel J. Bernstein	NSA
Structure of Algorithms	Feistel	Feistel	Substitution Permutation	Feistel	Substitution Permutation	ARX (Addition-Rotation-XOR)	ARX (Addition-Rotation-XOR)	Feistel
Block Cipher Used	Binary	Binary	Binary	Binary	Non-Binary	Binary	Binary (Stream Cipher variant)	Binary
Length of the Key	56 bits	112 bits, 168 bits	128 bits, 192 bits, 256 bits	32–448 bits	1–4096 set of integers	64, 72, 96, 128 bits	128, 256 bits	32, 48, 64, 96, 128 bits
Size of the Block	64 bits	64 bits	128 bits	64 bits	64 characters	32, 48, 64, 96, 128 bits	N/A (Stream Cipher)	32, 48, 64, 96, 128 bits
Value of Throughput	Lower than AES	Lower than DES	Lower than Blowfish	High	Lower than AES	Very High	Very High	High
Security Level	Sufficient security	Sufficient security	Good security	Good security	Highly secure	Moderate (controversial)	High	High
Speed of Encrypting Data	Slow	Very slow	Fast	Fast	Moderate	Very Fast	Very Fast	Fast
Effectiveness in Hardware/Software	Slow in software as well as hardware	Slow in software	Effective for software as well as hardware	Efficient for software	Efficient for software	Highly efficient in hardware	Highly efficient in software	Optimized for hardware
Prone to Threats/Attacks	Brute force	Known plaintext, Brute force, Chosen plaintext	Side-channel	Dictionary	No attacks as of now	Differential cryptanalysis	Timing attacks (mitigated in variants)	Differential cryptanalysis

TABLE 10 | Comparative analysis of key management approaches.

Key management approach	Case study/real-world deployment	Scalability and challenges	Industry trend
AWS KMS	Widely adopted by enterprises; integrates with 34 AWS services (S3, EC2, RDS, etc.). Used by financial and healthcare sectors for regulatory compliance	High scalability with automatic key rotation; challenges include potential performance degradation and AWS vendor lock-in	Increasing use in hybrid cloud environments, supporting multi-cloud security strategies
STYX	Implemented in Content Delivery Networks (CDNs) to ensure secure key distribution and data integrity. Used by media streaming services for secure content access	It provides high performance and low latency but requires a complex setup and integration with existing security policies	Adoption is growing in large-scale content distribution and multi-tenant cloud applications
BYOK (Bring Your Own Key) Models	Preferred by many enterprises for enhanced control over encryption keys, with support from AWS, Google Cloud, and Azure	Ensures compliance and data sovereignty but increases management complexity	Increasing demand due to regulatory requirements (GDPR, HIPAA), driving the development of user-managed key models

TABLE 11 | Future directions in key management systems.

Time horizon	Focus area	Objectives	Expected impact
Short-term (1–3 years)	Optimisation of Existing KMS	Improve latency, reduce overhead, and enhance the scalability of current systems	Improved performance for real-time applications (e.g., IoT, CDN)
Medium-term (3–5 years)	AI and Blockchain Integration	Introduce adaptive key management via ML and ensure decentralization via DLTs	Enhanced automation, trust, and security across cloud ecosystems
Long-term (5+ years)	Quantum-resistant KMS Protocols	Develop cryptographic schemes resilient to quantum attacks	Future-proof cloud security infrastructure

prehensive analysis of various cryptographic techniques applied in CC. The survey also discusses frameworks proposed for cloud platforms and how they intend to tackle multiple issues and challenges in CC. Although some novel possibilities are practical and may be used in a cloud system, others require additional testing and development.

6.4 | Practical Applications and Industry Trends

Three main key management approaches referred to as AWS KMS, STYX, and BYOK (Bring Your Own Key) models appear in Table 10. AWS KMS serves many organizations because of its ease of scale and integration with AWS services, especially in regulated sectors. Yet, it experiences system performance challenges and dependence on the vendor. STYX provides CDN users with secure content distribution, but the setup needs to be intricate. Users are choosing BYOK models more often because they enhance control and compliance, but these increased management needs drive implementation because of regulatory requirements to control keys in cloud setups.

7 | Conclusion and Future Directions

This SLR considers a list of 53 notable articles (from 2011 to 2024) on KM at the cloud scale. It examines the challenges and issues overlooking cryptographic KM from the perspective of architectures, frameworks, and algorithms widely used in various cryptographic operations. Our in-depth meta-analysis showed that KMS in CC is still an evolving and promising field of research.

The survey paper also focuses on developing a reliable cloud storage system while addressing security issues related to storing private data in the public cloud. One of the main techniques that CC systems should use to protect data is strong encryption with KM. Although there has been a lot of progress in this domain, this systematic survey also discusses considerable challenges that need to be addressed and resolved. Therefore, keeping in view the developments, inventions, and challenges discussed in this paper, future work can revolve around the given points:

1. Higher level of security that focuses on lower complexity and computation time, but a more robust encryption level.

2. Methods for deleting the encryption/decryption group keys in case of shared data to protect it after deletion.

3. Evaluation of crypto shredding technique integrated with different encryption algorithms such as DES encryption, Elgamal encryption scheme, etc.

4. Techniques to enhance the utilization of resources, adhere to role-based SLAs (service level agreements), and keep the system load-balanced, etc.

5. ECC (Elliptic-curve cryptography) and fuzzy logic can be used in combination to study how key-size-dependent reductions in key generation, encryption, and decryption time can be achieved in CC.

6. Optimizing latency in KMS through lightweight cryptographic protocols.

7. Developing adaptive key rotation strategies to balance security and performance.

8. Investigating user-centric KMS designs for enhanced privacy in multi-cloud settings.

KMS optimization during the following 3-year period will concentrate on improving latency and scalability, which benefit real-time applications (as shown in Table 11). For the period of 3–5 years, entities should combine AI computing with blockchain technologies to support adaptive oversight while strengthening security systems through automated processes. The development of quantum-resistant protocols remains the main objective for future years because these protocols will provide cloud infrastructure with long-term protection from quantum computing threats.

Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

References

1. D. Saxena, I. Gupta, J. Kumar, A. K. Singh, and X. Wen, "A Secure and Multiobjective Virtual Machine Placement Framework for Cloud Data Centers," *IEEE Systems Journal* 16 (2021): 3163–3174.

2. European Union Agency for Cybersecurity (ENISA), "Cloud Computing: Benefits, Risks, and Recommendations for Information Security." n.d., https://www.enisa.europa.eu/activities/risk-management/files/deliverables/cloud-computing-risk-assessment.

3. M. Y. Shakor, M. I. Khaleel, M. Safran, S. Alfarhood, and M. Zhu, "Dynamic AES Encryption and Blockchain Key Management: A Novel Solution for Cloud Data Security," *IEEE Access* 12 (2024): 26334–26343.

4. F. A. Alali and C.-L. Yeh, "Cloud Computing: Overview and Risk Analysis," *Journal of Information Systems* 26, no. 2 (2012): 13–33.

5. T. Probst, "Assessment and Analysis of Network Security Mechanisms in Virtual Cloud Infrastructures," (Master's Thesis, INP Toulouse: 2015), In French.

6. B. Hammi, "Towards Source Detection of Malicious Activities in Public Clouds: Application to Denial-Of-Service Attacks," (Doctoral dissertation, Troyes: 2015), In French.

7. D. Saxena and A. K. Singh, "Security-Embedded Dynamic Resource Allocation Model for the Cloud Data Center," *Electronics Letters* 56, no. 20 (2020): 1062–1065.

8. D. Saxena and A. K. Singh, "Proactive Autoscaling and Energy-Efficient VM Allocation Framework Using Online Multi-Resource Neural Network for the Cloud Data Center," *Neurocomputing* 426 (2021): 248–264.

9. D. Saxena and A. K. Singh, "OSC-MC: Online Secure Communication Model for Cloud Environment," *IEEE Communications Letters* 25 (2021): 2844–2848.

10. W. Book, "Cloud Computing Security: Analysis of Risks, Responses and Good Practices," (2010), In French.

11. Cigref, "Fundamentals of Cloud Computing — The Business Perspective," (2013), In French.

12. S. Chhabra and A. K. Singh, "A Probabilistic Model for Finding an Optimal Host Framework and Load Distribution in a Cloud Environment," *Procedia Computer Science* 125 (2018): 683–690.

13. N. Singh and A. K. Singh, "SQL-Injection Vulnerabilities Resolving Using Valid Security Tools in the Cloud," *Pertanika Journal of Science and Technology* 27, no. 1 (2019): 159–174.

14. J. Kumar, A. K. Singh, and R. Buyya, "Ensemble Learning-Based Predictive Framework for Virtual Machine Resource Request Prediction," *Neurocomputing* 397 (2020): 20–30.

15. S. Chhabra and A. K. Singh, "Dynamic Data Leakage Detection Model-Based Approach for MapReduce Computational Security in the Cloud," in *In 2016 Fifth International Conference on Eco-Friendly Computing and Communication Systems (ICECCS)* (IEEE, 2016), 13–19.

16. A. K. Singh and J. Kumar, "Secure and Energy-Aware Load Balancing Framework for Cloud Data Centre Networks," *Electronics Letters* 55, no. 9 (2019): 540–541.

17. A. S. Chauhan, D. Rani, A. Kumar, R. Gupta, and A. K. Singh, "A Survey on Privacy-Preserving Outsourced Data on the Cloud With Multiple Data Providers," in Proceedings of the International Conference on Innovative Computing and Communications (ICICC: 2020).

18. A. K. Singh and I. Gupta, "Online Information Leaker Identification Scheme for Secure Data Sharing," *Multimedia Tools and Applications* 79, no. 41 (2020): 31165–31182.

19. M. R. Deepika, S. Kumar, R. Gupta, and A. K. Singh, "A Review on Data Privacy Using Attribute-Based Encryption," in *Proceedings of the International Conference on Innovative Computing and Communications (ICICC)* (Springer, 2020).

20. S. Chhabra and A. K. Singh, "A Secure VM Allocation Scheme to Preserve Against the Co-Resident Threat," *International Journal of Web Engineering and Technology* 15, no. 1 (2020): 96–115.

21. S. Ahmad, M. Arif, S. Mehfuz, J. Ahmad, and M. Nazim, "Deep Learning-Based Cloud Security: Innovative Attack Detection and Privacy Focused Key Management," IEEE Transactions on Computers, https://doi.org/10.1109/TC.2025.3547150.

22. K. N. Kaur, I. Gupta, A. K. Singh, et al., "Digital Image Watermarking Using (2, 2) Visual Cryptography With DWT-SVD Based Watermarking," in *Computational Intelligence in Data Mining* (Springer, 2019), 77–86.

23. K. Kaur, I. Gupta, and A. K. Singh, "Data Leakage Prevention: E-Mail Protection via a Gateway," Journal of Physics: Conference Series, 933, 012013 (IOP Publishing: 2017).

24. S. Ahmad, S. Mehfuz, and J. Beg, "Cloud Security Framework and Key Management Services Collectively for Implementing DLP and IRM," *Materials Today: Proceedings* 62 (2022): 4828–4836, https://doi.org/10.1016/j.matpr.2022.03.420.

25. S. Ahmad, S. Mehfuz, and J. Beg, "Assessment on Potential Security Threats and Introducing Novel Data Security Model in a Cloud Environment," *Materials Today: Proceedings* 62 (2022): 4909–4915, https://doi.org/10.1016/j.matpr.2022.03.536.

26. S. Ahmad, S. Mehfuz, and J. Beg, "A Genetic Algorithm Based Key Management Approach for Enhancing Data Security in Cloud Environment," *International Journal of Advanced Trends in Computer Science and Engineering* 9, no. 5 (2020): 8943–8951.

27. X. Huang and R. Chen, "A Survey of Key Management Service in Cloud," in *In 2018 IEEE 9th International Conference on Software Engineering and Service Science (ICSESS)* (IEEE, 2018), 916–919, https://doi.org/10.1109/ICSESS.2018.8663805.

28. L. Probasco, "Townsend Security Data Privacy Blog: State of Encryption Key Management," (2020), https://info.townsendsecurity.com/state-of-encryption-key-management-2020.

29. X. Hu, J. Li, C. Wei, et al., "STYX: A Hierarchical Key Management System Oriented to Elastic Content Delivery Networks on Public Clouds," *IEEE Transactions on Dependable and Secure Computing* 99 (2019): 1, https://doi.org/10.1109/TDSC.2019.2919189.

30. A. Nag, M. M. Hassan, A. Das, et al., "Exploring the Applications and Security Threats of Internet of Things in the Cloud Computing Paradigm: A Comprehensive Study on the Cloud of Things," *Transactions on Emerging Telecommunications Technologies* 35, no. 4 (2024): e4897, https://doi.org/10.1002/ett.4897.

31. M. Campagna and S. Gueron, "Key Management Systems at the Cloud Scale," *Cryptography* 3, no. 23 (2019): 1–16, https://doi.org/10.3390/cryptography3030023.

32. L. Shakkeera, A. Saranya, and S. S. Y. Vali, "Secure Collaborative Key Management System for Mobile Cloud Data Storage," *International Journal of Engineering and Advanced Technology (IJEAT)* 8, no. 5S3 (2019): 214–225.

33. K. V. Pradeep and V. Vijayakumar, "Secure Key Management System in Cloud Environment for Client Data," *International Journal of Engineering and Advanced Technology (IJEAT)* 8, no. 5 (2019): 1490–1496.

34. C. Thirumalai, S. Mohan, and G. Srivastava, "An Efficient Public Key Secure Scheme for Cloud and IoT Security," *Computer Communications* 150 (2020): 634–643.

35. D. Shivaramakrishna and M. Nagaratna, "A Novel Hybrid Cryptographic Framework for Secure Data Storage in Cloud Computing: Integrating AES-OTP and RSA With Adaptive Key Management and Time-Limited Access Control," *Alexandria Engineering Journal* 84 (2023): 275–284.

36. M. Ben-Or, S. Goldwasser, and A. Wigderson, "Completeness Theorems for Non-Cryptographic Fault-Tolerant Distributed Computation," in *Proceedings of the 20th Annual ACM Symposium on Theory of Computing* (ACM, 1988), 1–10.

37. D. Chaum, C. Crépeau, and I. Damgård, "Multiparty Unconditionally Secure Protocols," in *Proceedings of the 20th Annual ACM Symposium on Theory of Computing* (ACM, 1988), 11–19.

38. O. Goldreich, S. Micali, and A. Wigderson, "How to Play Any Mental Game or a Completeness Theorem for Protocols With an Honest Majority," in *Proceedings of the Nineteenth Annual ACM Symposium on Theory of Computing (STOC)*, ed. A. V. Aho (Association for Computing Machinery, 1987), 218–229, ACM.

39. A. C.-C. Yao, "Protocols for Secure Computations (Extended Abstract)," in *Proceedings of the 23rd Annual Symposium on Foundations of Computer Science (FOCS)* (IEEE Computer Society, 1982), 160–164.

40. A. C.-C. Yao, "How to Generate and Exchange Secrets (Extended Abstract)," in *Proceedings of the 27th Annual Symposium on Foundations of Computer Science (FOCS)* (IEEE Computer Society, 1986), 162–167.

41. P. Bogetoft, D. L. Christensen, I. Damgård, et al., "Secure Multiparty Computation Goes Live," in *Financial Cryptography and Data Security (Lecture Notes in Computer Science)*, ed. R. Dingledine and P. Golle (Springer, 2009), 325–343.

42. C. Gentry, "Computing on Encrypted Data," in *Cryptology and Network Security (CANS) (Lecture Notes in Computer Science)*, vol. 5888, ed. J. A. Garay, A. Miyaji, and A. Otsuka (Springer, 2009), 477.

43. C. Gentry, "A Fully Homomorphic Encryption Scheme," (Ph.D. thesis, Stanford University: 2009), http://crypto.stanford.edu/craig.

44. R. Canetti, B. Riva, and G. N. Rothblum, "Refereed Delegation of Computation," *Information and Computation* 226 (2013): 16–36.

45. K.-M. Chung, Y. T. Kalai, and S. P. Vadhan, "Improved Delegation of Computation Using Fully Homomorphic Encryption," in *Proceedings of the 6th Innovations in Theoretical Computer Science Conference (ITCS)*, ed. O. Goldreich (Association for Computing Machinery, 2013), 483–501, ACM.

46. R. Gennaro, C. Gentry, and B. Parno, "Non-Interactive Verifiable Computing: Outsourcing Computation to Untrusted Workers," in *Proceedings of the 6th Innovations in Theoretical Computer Science Conference (ITCS)*, ed. O. Goldreich (Association for Computing Machinery, 2013), 465–482, ACM.

47. S. Kamara, P. Mohassel, and M. Raykova, "Outsourcing Multi-Party Computation," *IACR Cryptology ePrint Archive* 2011, no. 272 (2011).

48. Z. Brakerski, C. Gentry, and V. Vaikuntanathan, "(Leveled) Fully Homomorphic Encryption Without Bootstrapping," in *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference (ITCS)*, ed. S. Goldwasser (Association for Computing Machinery, 2012), 309–325, ACM.

49. S. Halevi and V. Shoup, "HElib: A Software Library Implementing Homomorphic Encryption," (2012), https://github.com/shaih/HElib.

50. N. Alon, H. Kaplan, M. Krivelevich, D. Malkhi, and J. P. Stern, "Scalable Secure Storage When Half the System Is Faulty," in *Automata, Languages and Programming (ICALP) (Lecture Notes in Computer Science)*, vol. 1853, ed. U. Montanari, J. D. P. Rolim, and E. Welzl (Springer, 2000), 576–587.

51. J. A. Garay, R. Gennaro, C. S. Jutla, and T. Rabin, "Secure Distributed Storage and Retrieval," *Theoretical Computer Science* 243, no. 1–2 (2000): 363–389.

52. S. Lakshmanan, M. Ahamad, and H. Venkateswaran, "Responsive Security for Stored Data," in *Proceedings of the 23rd International Conference on Distributed Computing Systems (ICDCS)* (IEEE Computer Society, 2003), 146.

53. H. Krawczyk, "Distributed Fingerprints and Secure Information Dispersal," in *Proceedings of the 12th Annual ACM Symposium on Principles of Distributed Computing (PODC)*, ed. J. Anderson and S. Toueg (Association for Computing Machinery, 1993), 207–218, ACM.

54. M. O. Rabin, "Efficient Dispersal of Information for Security, Load Balancing, and Fault Tolerance," *Journal of the ACM* 36, no. 2 (1989): 335–348.

55. B. Libert and M. Yung, "Adaptively Secure Forward-Secure Non-Interactive Threshold Cryptosystems," in *Information Security and Cryptology (Inscrypt) (Lecture Notes in Computer Science)*, vol. 7537, ed. C. Wu, M. Yung, and D. Lin (Springer, 2011), 1–21.

56. G. Lin, H. Hong, and Z. Sun, "A Collaborative Key Management Protocol in Ciphertext-Policy Attribute-Based Encryption for Cloud Data Sharing," *IEEE Access* 5 (2017): 9464–9475.

57. S. Easwarmoorthy, F. Sophia, and A. Karrothu, "An Efficient Key Management Infrastructure for Personal Health Records in Cloud," in *WiSPNET* (IEEE, 2016), 1651–1657.

58. Y. S. Abdulsalam, J. Bouamama, Y. Benkaouz, and M. Hedabou, "Decentralized SGX-Based Cloud Key Management," in *International Conference on Network and System Security* (Springer, 2023), 327–341.

59. L. Keuninckx, M. C. Soriano, I. Fischer, C. R. Mirasso, R. M. Nguimdo, and G. Van der Sande, "Encryption Key Distribution via Chaos Synchronization," *Scientific Reports* 7 (2017): 1–14.

60. J. Han and J. Wang, "An Enhanced Key Management Scheme for LoRaWAN," *Cryptography* 2, no. 4 (2018): 1–12.

61. H. Kwon, C. Hahn, D. Koo, and J. Hur, "Scalable and Reliable Key Management for Secure De-Duplication in Cloud Storage. In IEEE International Conference on Cloud Computing," (2017), 391–398.

62. D. Sitaram, S. Harwalkar, U. Simha, S. Iyer, and S. Jha, "Standards-Based Integration of Advanced Key Management Capabilities With OpenStack. In IEEE International Conference on Cloud Computing in Emerging Markets," (2016), 98–103.

63. X. Tang, C. Guo, K. K. R. Choo, X. Jiang, and Y. Liu, "A Secure and Lightweight Cloud Data Deduplication Scheme With Efficient Access Control and Key Management," *Computer Communications* 222 (2024): 209–219.

64. P. Bajpai, A. K. Sood, and R. Enbody, *A Key-Management-Based Taxonomy for Ransomware* (In APWG Symposium on Electronic Crime Research, 2018).

65. A. Elhadad, *Data Sharing Using Proxy Re-Encryption Based on DNA Computing* (Springer, 2019).

66. Q. Gu and M. Potkonjak, *Efficient and Secure Group Key Management in IoT Using Multistage Interconnected PUF* (IEEE, 2018).

67. N. Saad, "A Trusted Way for Encryption Key Management in Cloud Computing," in *International Conference on Advanced Information Technology* (Springer, 2017).

68. M. R. Alshammari and K. M. Elleithy, "Efficient and Secure Key Distribution Protocol for Wireless Sensor Networks," *Sensors* 18, no. 11 (2018): 1–25.

69. R. Dowsley, M. Gabel, G. Hubsch, and G. Schiefer, "A Distributed Key Management Approach," in *IEEE International Conference on Cloud Computing Technology and Science (CloudCom)* (IEEE, 2016), 509–514.

70. V. Gurusamy, S. Kannan, and T. M. Mahajan, "Cryptographic Tree and Its Key Management for Securing Outsourced Data in the Cloud," *International Journal of Advanced Research in Computer Science and Software Engineering* 7, no. 5 (2017): 255–259.

71. Y. Pachipala and J. Alzubi, "Managing the Cloud Storage Using Deduplication and Secured Fuzzy Keyword Search for Multiple," *International Journal of Pure and Applied Mathematics* 118, no. 14 (2018): 563–565.

72. F. I. Kandah, O. Nichols, and L. Yang, "Efficient Key Management for Big Data Gathering in Dynamic Sensor Networks," in *Workshop on Computing, Networking and Communications (CNC)* (IEEE, 2017), 1–5.

73. X. W. Li, D. Q. Yang, B. X. Chen, et al., "Two-Factor Authenticated Key Agreement Protocol Based on Biometric Feature and Password," *Journal of Communication* 38, no. 7 (2017): 89–95.

74. C. Ge, W. Susilo, Z. Liu, J. Xia, P. Szalachowski, and F. Liming, "Secure Keyword Search and Data Sharing Mechanism for Cloud Computing," 2020 (2020), IEEE Transactions on Dependable and Secure Computing.

75. A. Irshad, S. A. Chaudhry, M. Shafiq, M. Usman, M. Asif, and A. Ghani, "A Provable and Secure Mobile User Authentication Scheme for Mobile Cloud Computing Services," *International Journal of Communication Systems* 32, no. 14 (2019): e3980, https://doi.org/10.1002/dac.3980.

76. W. Liu, X. Wang, W. Peng, and Q. Xing, "Center-Less Single Sign-On With Privacy-Preserving Remote Biometric-Based ID-MAKA Scheme for Mobile Cloud Computing Services," *IEEE Access* 7 (2019): 137770–137783.

77. M. Vivekanandan, V. N. Sastry, and U. S. Reddy, "Biometric Based User Authentication Protocol for Mobile Cloud Environment," in *In 2019 IEEE 5th International Conference on Identity, Security, and Behavior Analysis (ISBA)* (IEEE, 2019), 1–6.

78. X. Yi, "Identity-Based Fault-Tolerant Conference Key Agreement," *IEEE Transactions on Dependable and Secure Computing* 1, no. 3 (2004): 170–178.

79. C. S. Jang, D. G. Lee, J. W. Han, and J. H. Park, "Hybrid Security Protocol for Wireless Body Area Networks," *Wireless Communications and Mobile Computing* 11, no. 2 (2011): 277–288.

80. D. He, S. Zeadally, N. Kumar, and W. Wu, "Efficient and Anonymous Mobile User Authentication Protocol Using Self-Certified Public Key Cryptography for Multi-Server Architectures," *IEEE Transactions on Information Forensics and Security* 11, no. 9 (2016): 2052–2064.

81. H. Tan, Y. Song, S. Xuan, S. Pan, and I. Chung, "Secure D2D Group Authentication Employing Smartphone Sensor Behavior Analysis," *Symmetry* 11, no. 8 (2019): 969, https://doi.org/10.3390/sym11080969.

82. K. Gai, I. Guo, L. Zhu, and S. Yu, "Blockchain Meets Cloud Computing: A Survey," (2019), IEEE Communication Surveys and Tutorials, https://doi.org/10.1109/COMST.2020.2989392.

83. J. Huang and J. Yi, "The Key Security Management Scheme of Cloud Storage Based on Blockchain and Digital Twins," *Journal of Cloud Computing* 13, no. 1 (2024): 15, https://doi.org/10.1186/s13677-023-00587-4.

84. K. S. Kumar, R. Gupta, S. Ramesh, and H. Mehta, *A Secure and Efficient Blockchain and Distributed Ledger Technology-Based Optimal Resource Management in Digital Twin Beyond 5G Networks Using Hybrid Energy Valley and Levy Flight Distributer Optimization Algorithm* (IEEE Access, 2024).

85. P. Bajpai and R. Enbody, "Know Thy Ransomware Response: A Detailed Framework for Devising Effective Ransomware Response Strategies," *Digital Threats: Research and Practice* 4, no. 4 (2023): 1–19.

86. S. Velmurugan, M. Prakash, S. Neelakandan, et al., "Provably Secure Data Selective Sharing Scheme With Cloud-Based Decentralized Trust Management Systems," *Journal of Cloud Computing* 13 (2024): 86, https://doi.org/10.1186/s13677-024-00634-8.

87. Y. S. Abdulsalam, J. Bouamama, Y. Benkaouz, and M. Hedabou, "Decentralized SGX-Based Cloud Key Management," in *Network and System Security. NSS 2023. Lecture Notes in Computer Science*, vol. 13983, ed. S. Li, M. Manulis, and A. Miyaji (Springer, 2023), https://doi.org/10.1007/978-3-031-39828-5_18.

88. X. Tang, Y. Wang, H. Zhang, T. Li, X. Zhao, and W. Zhao, "LLM-Assisted Reinforcement Learning: Leveraging Lightweight Large Language Model Capabilities for Efficient Task Scheduling in Multi-Cloud Environment," (2024), IEEE Transactions on Consumer Electronics, https://doi.org/10.1109/TCE.2024.3524612.

89. A. Susanto, A. H. Fathulloh, and A. Fitriyani, "Comparative Analysis of Key Management Service Performance on AWS, Google Cloud, and Oracle Cloud With Performance Testing," in *In 2023 11th International Conference on Cyber and IT Service Management (CITSM)* (IEEE, 2023), 1–6.

90. L. Xiong, J. Wang, L. Yu, N. Xiong, and H. Wu, "An Efficient Privacy-Preserving Access Control Scheme for Cloud Computing Services," (2025), IEEE Transactions on Consumer Electronics, https://doi.org/10.1109/TCE.2025.3534833.

91. Z. Hou, T. Pei, M. Li, K. Wei, Y. Chen, and S. Cao, "A Fault-Tolerant Group Key Management Scheme for the Internet of Things Based on Multi-Layer Blockchain," *IEEE Internet of Things Journal* 12 (2025): 18004–18018, https://doi.org/10.1109/JIOT.2025.3540171.

92. Q. Zhang, S. Qian, J. Cui, H. Zhong, F. Wang, and D. He, "Blockchain-Based Privacy-Preserving Deduplication and Integrity Auditing in Cloud Storage," *IEEE Transactions on Computers* 74 (2025): 1717–1729, https://doi.org/10.1109/TC.2025.3540670.

93. R. Kumar, A. Gupta, R. Singh, V. Sharma, and A. Singh, "Secure and Cost-Effective Key Management Scheme for the Internet of Things-Supported WSN," in *Uncertainty in Computational Intelligence-Based Decision Making* (Academic Press, 2025), 277–292, https://doi.org/10.1016/B978-0-443-21475-2.00012-6.

94. Y. Elkhatib, "Mapping Cross-Cloud Systems: Challenges and Opportunities," in *Proceedings of the 8th USENIX Conference on Hot Topics in Cloud Computing* (USENIX Association, 2016), 77–83.

95. S. A. Khah, A. Barati, and H. Barati, "A Dynamic and Multi-Level Key Management Method in Wireless Sensor Networks (WSNs)," *Computer Networks* 236 (2023): 109997, https://doi.org/10.1016/j.comnet.2023.109997.

96. P. Alimoradi, A. Barati, and H. Barati, "A Hierarchical Key Management and Authentication Method for Wireless Sensor Networks," *International Journal of Communication Systems* 35, no. 6 (2022): e5076, https://doi.org/10.1002/dac.5076.

97. H. Barati, "A Hierarchical Key Management Method for Wireless Sensor Networks," *Microprocessors and Microsystems* 90 (2022): 104489, https://doi.org/10.1016/j.micpro.2022.104489.

98. S. Doostani, H. Barati, and A. Barati, "A Lightweight Hierarchical Method for Improving Security in the Internet of Things Using Fuzzy Logic," *Concurrency and Computation: Practice and Experience* 36, no. 6 (2024): e7959, https://doi.org/10.1002/cpe.7959.

99. S. Ahmad, S. Mehfuz, and J. Beg, "Securely Work From Home With CASB Policies Under COVID-19 Pandemic: A Short Review," in *In 2020 9th International Conference on System Modeling and Advancement in Research Trends (SMART)* (IEEE, 2020), 109–114, https://doi.org/10.1109/SMART50582.2020.9337121.

100. A. Lamsal, A. Paul, A. Buchade, J. Nalavade, and R. Bidwe, "Implementing Secure Key Management and Distribution in Cloud Environment," in *In 2023 IEEE International Conference on Blockchain and Distributed Systems Security (ICBDS)* (IEEE, 2023), 1–10.

101. H. Ning, H. Liu, and L. T. Yang, "Aggregated-Proof Based Hierarchical Authentication Scheme for the Internet of Things," *IEEE Transactions on Parallel and Distributed Systems* 26, no. 3 (2015): 657–667.

102. T. Hu, J. Wang, G. Zhao, and X. Long, "An Improved Mutual Authentication and Key Update Scheme for Multi-Hop Relay in the Internet of Things," (2012), In Proceedings of the 7th IEEE Conference on Industrial Electronics and Applications (ICIEA). 1024–1029.

103. M. Wazid, A. K. Das, N. Kumar, and A. V. Vasilakos, "Design of Secure Key Management and User Authentication Scheme for Fog Computing Services," *Future Generation Computer Systems* 91 (2019): 475–492, https://doi.org/10.1016/j.future.2018.09.017.

104. S. A. Khan, R. K. Aggarwal, and S. Kulkarni, "Encryption Schemes of Cloud Computing: A Review," in *In 5th International Conference on Advanced Computing and Communication Systems (ICACCS)* (IEEE, 2019), 18–23.

105. J. Zhang, B. Chen, Y. Zhao, X. Cheng, and F. Hu, "Data Security and Privacy-Preserving in Edge Computing Paradigm: Survey and Open Issues," *IEEE Access* 6 (2018): 18209–18237.

106. K. Sudharani and P. N. K. Sakthivel, "A Secure Encryption Scheme Based on Certificateless Proxy Signature," *International Journal of Cloud Computing* 7, no. 1 (2018): 40–64.

107. J. Clough, *Principles of Cybercrime* (Cambridge University Press, 2010), 5–8.

108. E. Willems, *Cyberdanger: Understanding and Guarding Against Cybercrime* (Springer, 2019), 187.

109. M. M. Alshabibi, A. K. Bu dookhi, and M. M. Hafizur Rahman, "Forensic Investigation, Challenges, and Issues of Cloud Data: A Systematic Literature Review," *Computer* 13, no. 8 (2024): 213.

110. H. S. Brar and G. Kumar, "Cybercrimes: A Proposed Taxonomy and Challenges," *Journal of Computer Networks and Communications* 2018 (2016): 1798659, https://doi.org/10.1155/2018/1798659.

111. I. A. Mohammed, "Intelligent Authentication for Identity and Access Management: A Review Paper," *International Journal of Management, IT and Engineering (IJMIE)* 3, no. 1 (2013): 696–705, https://www.ijmra.us/project%20doc/IJMIE_JANUARY2013/IJMIEJan13Ishaq.pdf.

112. I. A. Mohammed, "Economics of Identity and Access Management: Providing Decision Support for Investments," *International Journal of Management, IT and Engineering (IJMIE)* 4, no. 2 (2014): 540–549, https://www.ijmra.us/project%20doc/2014/IJMIE_FEBRUARY2014/IJMIEFeb14-Ishaq.pd.

113. I. M. Khalil, A. Khreishah, and M. Azeem, "Cloud Computing Security: A Survey," *Computers* 3 (2014): 1–35.

114. D. S. Wall, "Towards a Conceptualization of Cloud (Cyber) Crime," in *Human Aspects of Information Security, Privacy and Trust*, ed. T. Tryfonas (Springer, 2017), 533–535.

115. S. N. Mithunzi, M. Mutemwa, A. Ismail, and C. Mutemwa, "Cloud Computing Security Taxonomy: From Atomistic to Holistic View," (2019), Future Generation Computer Systems, Advance Online Publication, https://doi.org/10.1016/j.future.2019.11.013.

116. R. Kumar and R. Goyal, "On Cloud Security Requirements, Threats, Vulnerabilities, and Countermeasures: A Survey," *Computer Science Review* 33 (2019): 1–48.

117. P. K. Senyo, E. Addae, and R. Boateng, "Cloud Computing Research: A Review of Research Themes, Frameworks, Methods, and Future Research Directions," *International Journal of Information Management* 38 (2018): 128–139.

118. M. De Donno, N. Dragoni, and A. Giaretta, "Cyber-Storms Come From Cloud: Security of Cloud Computing in the IoT Era," *Future Internet* 11, no. 127 (2019): 1–30.

119. S. Singh, Y. S. Jeong, and J. H. Park, "A Survey of Cloud Computing Security: Issues, Threats, and Solutions," *Journal of Network and Computer Applications* 75 (2016): 200–222.

120. D. A. B. Fernandes, L. F. B. Soares, J. V. Gomes, M. M. Freire, and P. R. M. Inácio, "Security Issues in Cloud Computing Environments," *International Journal of Information Security* 13 (2014): 113–170.

121. C. Modi, D. Patel, B. Borisaniya, A. Patel, and M. Rajarajan, "A Survey on Security Issues and Solutions at Different Layers of Cloud Computing," *Journal of Supercomputing* 63 (2013): 561–592.

122. D. Zissis and D. Lekkas, "Addressing Cloud Computing Security," *Future Generation Computer Systems* 28 (2012): 583–592.

123. J. Yuan and S. Yu, "Privacy-Preserving Back-Propagation Neural Network Learning Made Practical With Cloud Computing," *IEEE Transactions on Parallel and Distributed Systems* 25, no. 1 (2013): 212–221.

124. Q. Zhang, L. T. Yang, and Z. Chen, "Privacy-Preserving Deep Computation Model on the Cloud for Big Data Feature Learning," *IEEE Transactions on Computers* 65, no. 5 (2015): 1351–1362.

125. R. Yonetani, V. N. Boddeti, K. M. Kitani, and Y. Sato, "Privacy-Preserving Visual Learning Using Doubly Permuted Homomorphic Encryption," (2017), In Proceedings of the IEEE International Conference on Computer Vision, 2040–2050.

126. P. Li, J. Li, Z. Huang, et al., "Multi-Key Privacy-Preserving Deep Learning in Cloud Computing," *Future Generation Computer Systems* 74 (2017): 76–85.

127. P. Li, J. Li, Z. Huang, C.-Z. Gao, W.-B. Chen, and K. Chen, "Privacy-Preserving Outsourced Classification in Cloud Computing," *Cluster Computing* 21, no. 1 (2018): 277–286.

128. X. Ma, J. Ma, H. Li, Q. Jiang, and S. Gao, *PDLM: Privacy-Preserving Deep Learning Model on the Cloud With Multiple Keys* (IEEE Transactions on Services Computing, 2018).

129. T. Li, Z. Huang, P. Li, Z. Liu, and C. Jia, "Outsourced Privacy-Preserving Classification Service Over Encrypted Data," *Journal of Network and Computer Applications* 106 (2018): 100–110.

130. C.-Z. Gao, Q. Cheng, P. He, W. Susilo, and J. Li, "Privacy-Preserving Naive Bayes Classifiers Secure Against the Substitution-Then-Comparison Attack," *Information Sciences* 444 (2018): 72–88.

131. L. Qi, X. Zhang, W. Dou, and Q. Ni, "A Distributed Locality-Sensitive Hashing-Based Approach for Cloud Service Recommendation From Multi-Source Data," *IEEE Journal on Selected Areas in Communications* 35, no. 11 (2017): 2616–2624.

132. T. Wang, J. Zhou, X. Chen, G. Wang, A. Liu, and Y. Liu, "A Three-Layer Privacy-Preserving Cloud Storage Scheme Based on Computational Intelligence in Fog Computing," *IEEE Transactions on Emerging Topics in Computational Intelligence* 2, no. 1 (2018): 3–12.

133. Q. Huang and H. Li, "An Efficient Public-Key Searchable Encryption Scheme Secures Against Inside Keyword Guessing Attacks," *Information Sciences* 403 (2017): 1–14.

134. Z. Fu, L. Xia, X. Sun, A. X. Liu, and G. Xie, "Semantic-Aware Searching Over Encrypted Data for Cloud Computing," *IEEE Transactions on Information Forensics and Security* 13, no. 9 (2018): 2359–2371.

135. Y. Chen and Q. Yan, "Privacy-Preserving Diffusion Model Using Homomorphic Encryption," (2024), arXiv preprint arXiv:2403.05794.

136. Z. Fu, X. Wu, C. Guan, X. Sun, and K. Ren, "Toward Efficient Multi-Keyword Fuzzy Search Over Encrypted Outsourced Data With Accuracy Improvement," *IEEE Transactions on Information Forensics and Security* 11, no. 12 (2016): 2706–2716.

137. C. Guo, X. Chen, Y. Jie, F. Zhangjie, M. Li, and B. Feng, "Dynamic Multi-Phrase Ranked Search Over Encrypted Data With Symmetric Searchable Encryption," (2017), IEEE Transactions on Services Computing.

138. N. Cao, C. Wang, M. Li, K. Ren, and W. Lou, "Privacy-Preserving Multi-Keyword Ranked Search Over Encrypted Cloud Data," *IEEE Transactions on Parallel and Distributed Systems* 25, no. 1 (2013): 222–233.

139. R. Shokri and V. Shmatikov, "Privacy-Preserving Deep Learning. In Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security," (2015), 1310–1321.

140. S. Samet and A. Miri, "Privacy-Preserving Back-Propagation and Extreme Learning Machine Algorithms," *Data and Knowledge Engineering* 79 (2012): 40–61.

141. Q. Wang, S. Hu, M. Du, J. Wang, and K. Ren, "Learning Privately: Privacy-Preserving Canonical Correlation Analysis for Cross-Media Retrieval," in *IEEE INFOCOM 2017—IEEE Conference on Computer Communications* (IEEE, 2017), 1–9.

142. A. Bansal, T. Chen, and S. Zhong, "Privacy-Preserving Back-Propagation Neural Network Learning Over Arbitrarily Partitioned Data," *Neural Computing and Applications* 20, no. 1 (2011): 143–150.

143. T. Chen and S. Zhong, "Privacy-Preserving Backpropagation Neural Network Learning," *IEEE Transactions on Neural Networks* 20, no. 10 (2009): 1554–1564.

144. S. Fletcher and M. Z. Islam, "Differentially Private Random Decision Forests Using Smooth Sensitivity," *Expert Systems With Applications* 78 (2017): 16–31.

145. J. Sharma, D. Kim, A. Lee, and D. Seo, "On Differential Privacy-Based Framework for Enhancing User Data Privacy in a Mobile Edge Computing Environment," *IEEE Access* 9 (2021): 38107–38118.

146. P. Li, T. Li, H. Ye, J. Li, X. Chen, and Y. Xiang, "Privacy-Preserving Machine Learning With Multiple Data Providers," *Future Generation Computer Systems* 87 (2018): 341–350.

147. I. Gupta, R. Gupta, A. K. Singh, and R. Buyya, "MLPAM: A Machine Learning and Probabilistic Analysis-Based Model for Preserving Security and Privacy in a Cloud Environment," *IEEE Systems Journal* 99 (2020): 1–12.

148. T. Li, J. Li, Z. Liu, P. Li, and C. Jia, "Differentially Private Naive Bayes Learning Over Multiple Data Sources," *Information Sciences* 444 (2018): 89–104.

149. X. Liu, Q. Li, T. Li, and D. Chen, "Differentially Private Classification With Decision Tree Ensemble," *Applied Soft Computing* 62 (2018): 807–816.

150. Z. Wang, X. Pang, Y. Chen, et al., "Privacy-Preserving Crowd-Sourced Statistical Data Publishing With an Untrusted Server," *IEEE Transactions on Mobile Computing* 18, no. 6 (2018): 1356–1367.

151. A. Hassan, R. Hamza, H. Yan, and P. Li, "An Efficient Outsourced Privacy-Preserving Machine Learning Scheme With Public Verifiability," *IEEE Access* 7 (2019): 146322–146330.

152. B. Jayaram, T. Suguna, S. Fairooz, and S. Vimal, "Secure Data Reduplication System With Efficient and Reliable Multi-Key Management in Cloud Storage," in *In 2023 3rd International Conference on Innovative Practices in Technology and Management (ICIPTM)* (IEEE, 2023), 1–6.

153. Gartner, "Gartner Magic Quadrant for Cloud Service Provider," (2017).

154. C. McKinnon and Forrester, "The Forrester Wave: Enterprise Content Management—Business Content Services 2017," 2017.

155. S. Ahmad, S. Mehfuz, and J. Beg, "An Efficient and Secure Key Management With the Extended Convolutional Neural Network for Intrusion Detection in Cloud Storage," *Concurrency and Computation: Practice and Experience* 35, no. 23 (2023): e7806, https://doi.org/10.1002/cpe.7806.

156. F. Liu, J. Tong, J. Mao, et al., "NIST Cloud Computing Reference Architecture (NIST SP 500-292)," (2011), National Institute of Standards and Technology, http://www.nist.gov/customcf/get_pdf.cfm?pub_id=909505.

157. L. Badger, D. Berstein, R. Bohn, et al., "US Government Cloud Computing Technology Roadmap (NIST SP 500-293)," (2011), NIST, http://www.nist.gov/itl/cloud/upload/SP_500_293_volumeI-2.pdf.

158. L. Badger, R. Bohn, S. Chu, et al., "US Government Cloud Computing Technology Roadmap (NIST SP 500-293)," (2011), NIST, http://www.nist.gov/itl/cloud/upload/SP_500_293_volumeII.pdf.

159. F. Bracci, A. Corradi, and L. Foschini, *Database Security Management for Healthcare SaaS in the Amazon AWS Cloud* (IEEE Computer, 2012).

160. S. Ahmad, S. Mehfuz, and J. Beg, "Hybrid Cryptographic Approach to Enhance the Mode of Key Management System in Cloud Environment," *Journal of Supercomputing* 79, no. 7 (2023): 7377–7413.

161. S. Ahmad, S. Mehfuz, S. Urooj, and N. Alsubaie, "Machine Learning-Based Intelligent Security Framework for Secure Cloud Key Management," *Cluster Computing* 27 (2024): 1–27.

162. Securosis, "Understanding and Selecting a Database Encryption or Tokenization Solution," n.d., http://securosis.com.

163. W. Jansen and T. Grance, *Guidelines on Security and Privacy in Public Cloud Computing (NIST SP 800–144)* (Department of Commerce, 2011), https://csrc.nist.gov/publications/nistpubs/800-144/SP800-144.pdf.

164. E. Consulting, "Cloud Key Management Services: Advantages and Disadvantages," n.d., https://encryptionconsulting.com.

165. R. S. Hirschprung, "Separating the Siamese Twins: A Proposed Methodology for Differentiating Between Privacy and Security," *Behaviour and Information Technology* 43, no. 12 (2024): 2945–2965.

166. M. A. Darwish, E. A. Markatou, and G. Smaragdakis, "Provable co-Owned Data Deletion With Zero-Residuals and Verifiability in Multi-Cloud Environment. In Proceedings of the 18th European Workshop on Systems Security," (2025), 77–83.

167. A. W. Services, "AWS KMS Common Errors," n.d., https://docs.aws.amazon.com/kms/latest/APIReference/CommonErrors.html.

168. Cloud Security Alliance, "Key Management in Cloud Services," n.d., https://cloudsecurityalliance.org/.

169. K. Pradeep and V. Vijayakumar, "Survey on the Key Management for Securing the Cloud," *Procedia Computer Science* 50 (2015): 115–121, https://doi.org/10.1016/j.procs.2015.04.072.

170. M. Faisal, I. Ali, M. S. Khan, J. Kim, and S. M. Kim, "Cyber Security and Key Management Issues for Internet of Things: Techniques, Requirements, and Challenges," *Security and Privacy* 2020 (2020): 1–9, https://doi.org/10.1155/2020/6619498.

171. I. Qiqieh, J. Alzubi, and O. Alzubi, "DNA Cryptography-Based Security Framework for Health-Cloud Data," *Computing* 107, no. 1 (2025): 35.

172. M. Mushtaq, J. Sapie, D. Abdulkadir, P. Zahraddeen, N. Shakir, and M. A. Deris, "A Survey on the Cryptographic Encryption Algorithms," *International Journal of Advanced Computer Science and Applications* 8, no. 11 (2017): 333–344, https://doi.org/10.14569/IJACSA.2017.081141.